

CSE4203: Computer Graphics
Lecture – 6 (part - B)
Graphics Pipeline

Chapter 8

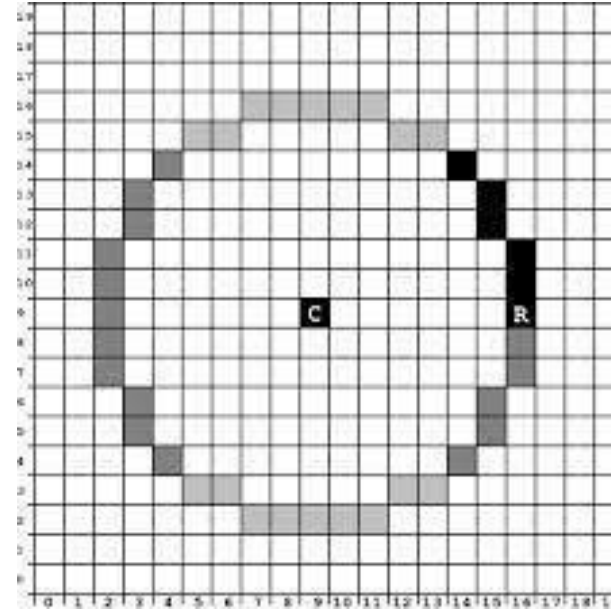
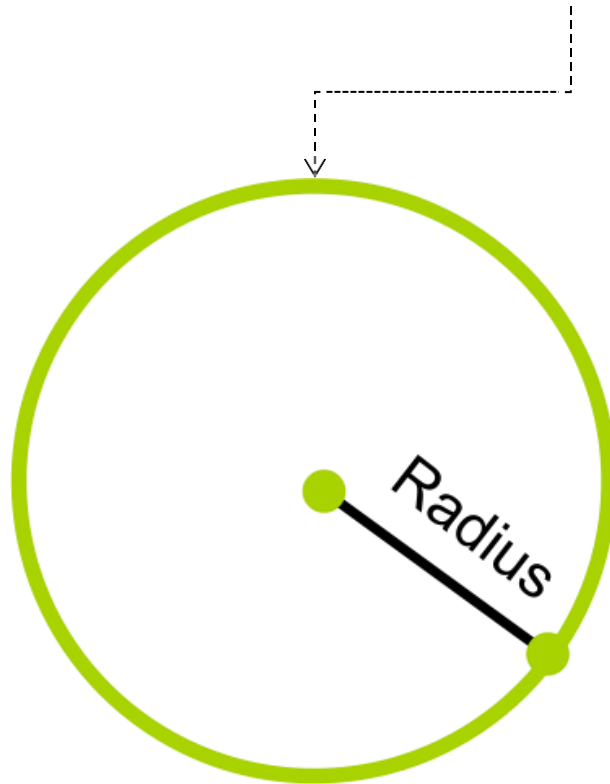
Outline

- Bresenham's Circle Drawing Algorithm

Assumptions

Given,
Radius R

circumference

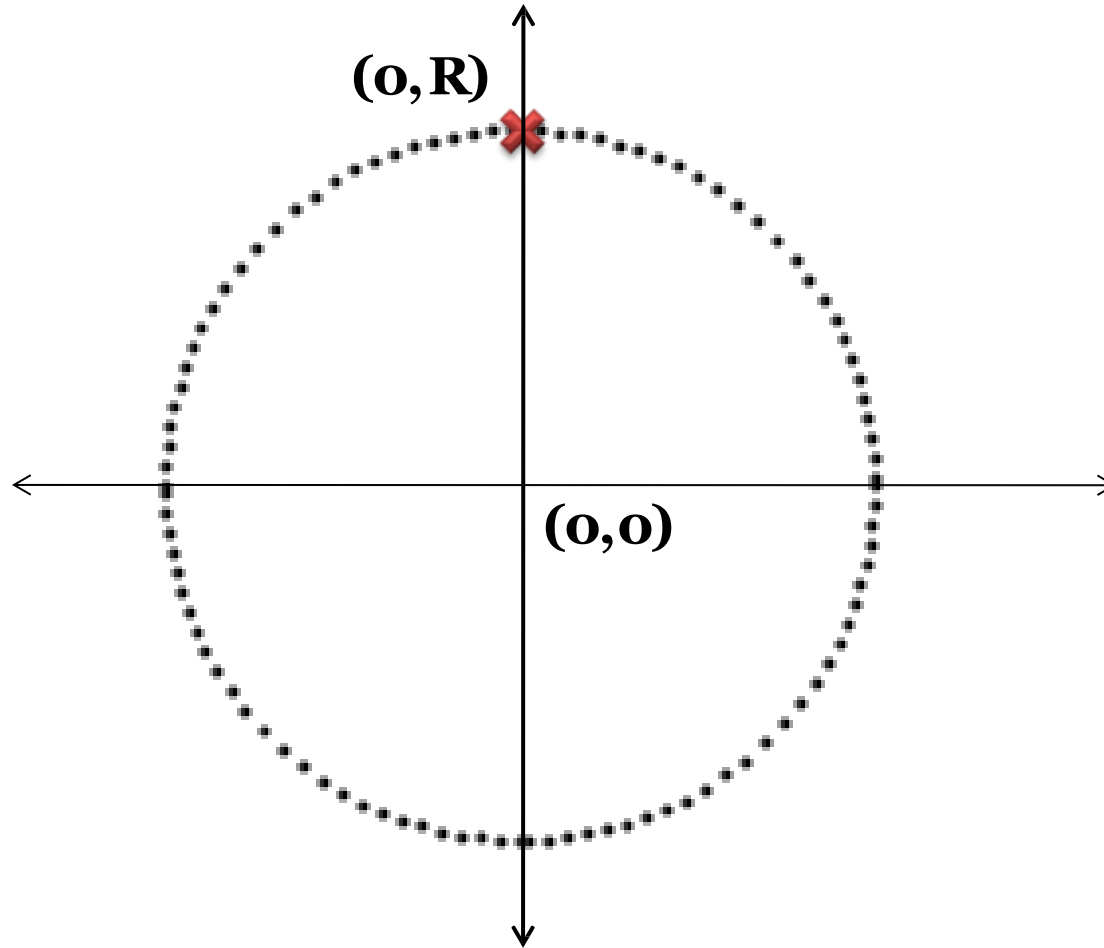


We have to develop an algorithm that generates this circumference

Assumptions

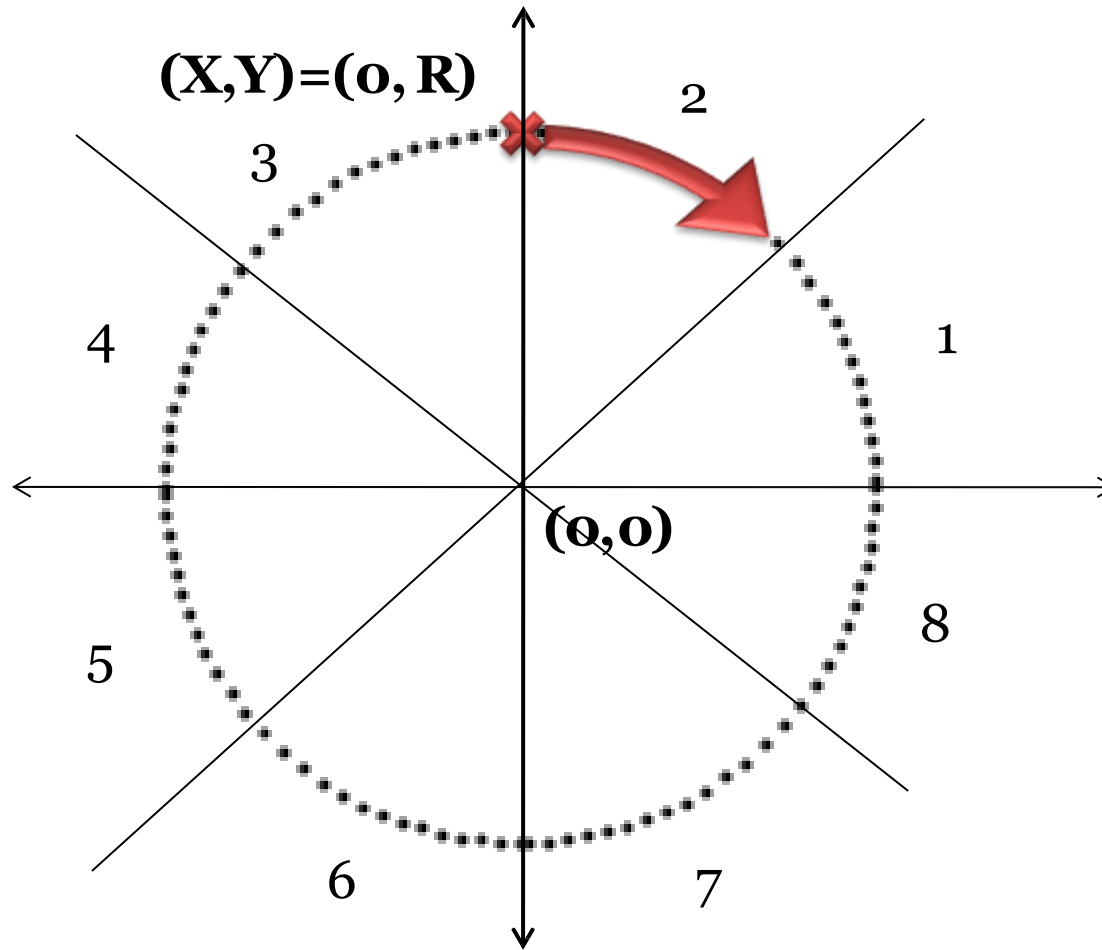
The first pixel of the circumference is plotted on $(0, R)$

Given,
Radius R



Observation

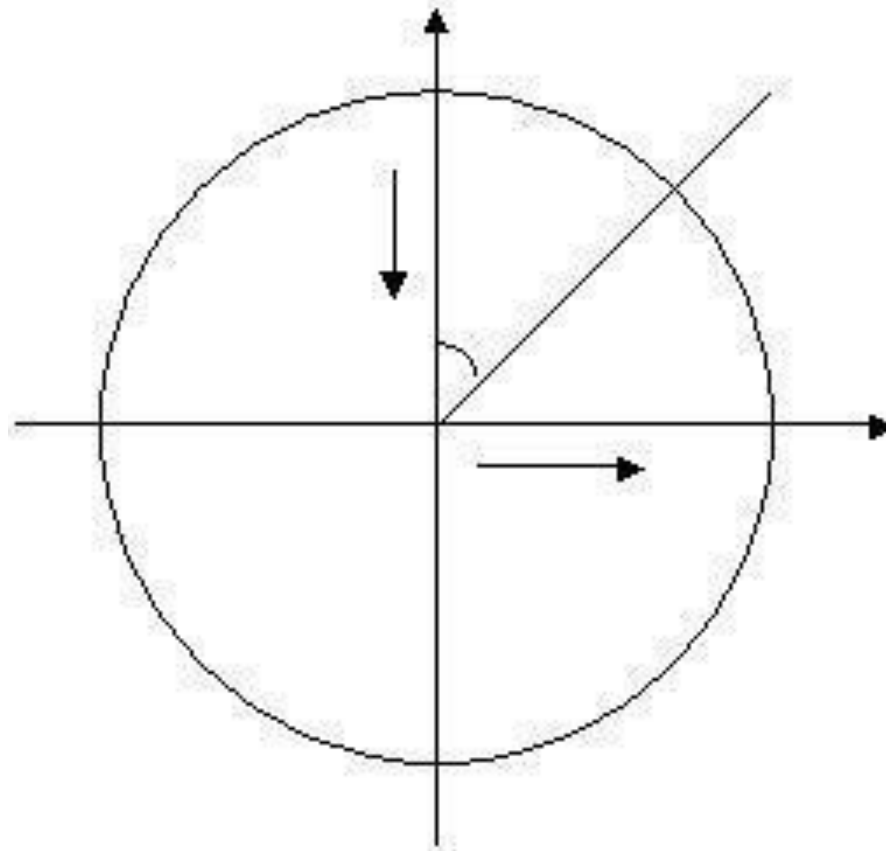
The first pixel of the circumference is plotted on $(0, R)$
Then the plotting of next pixels starts clock-wise....

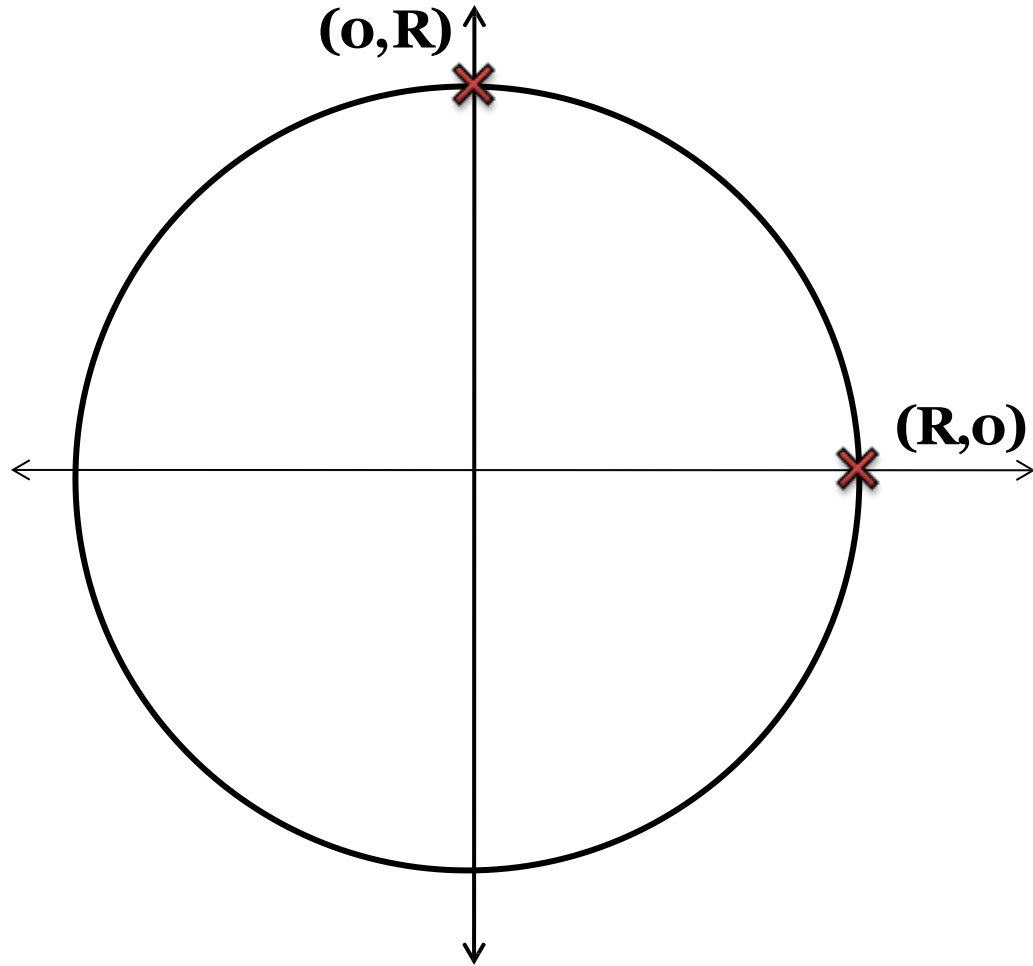


That means the plotting starts from $(0,R)$ and moving into the 2nd Octant

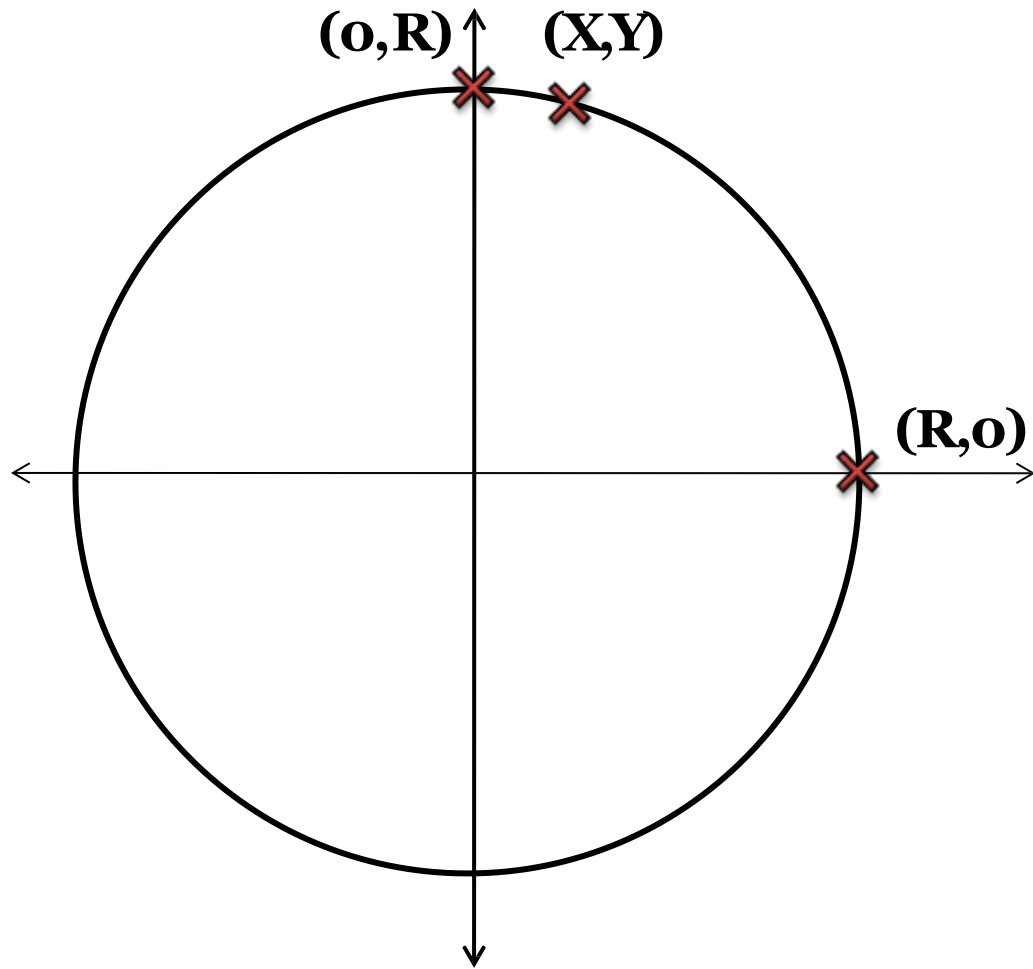
Observation

while moving through the 2nd octant, the X value is increasing and Y value is decreasing

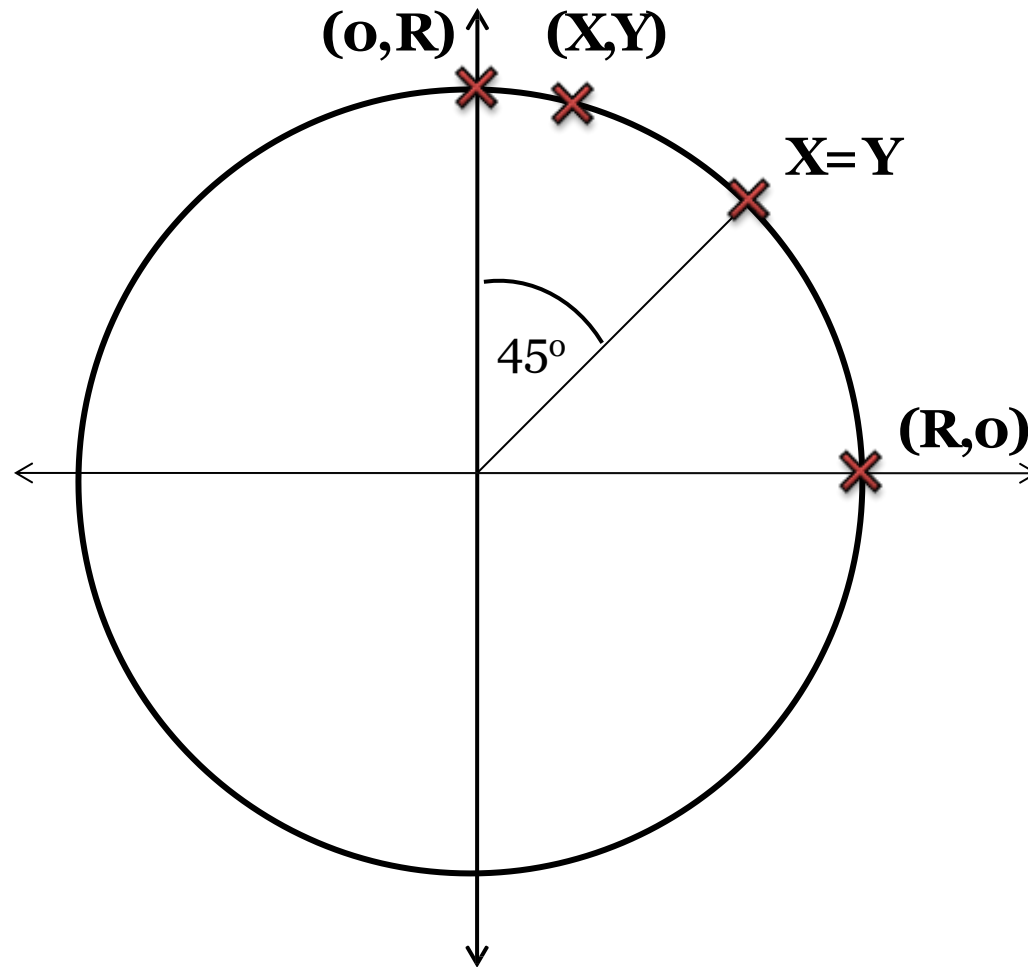




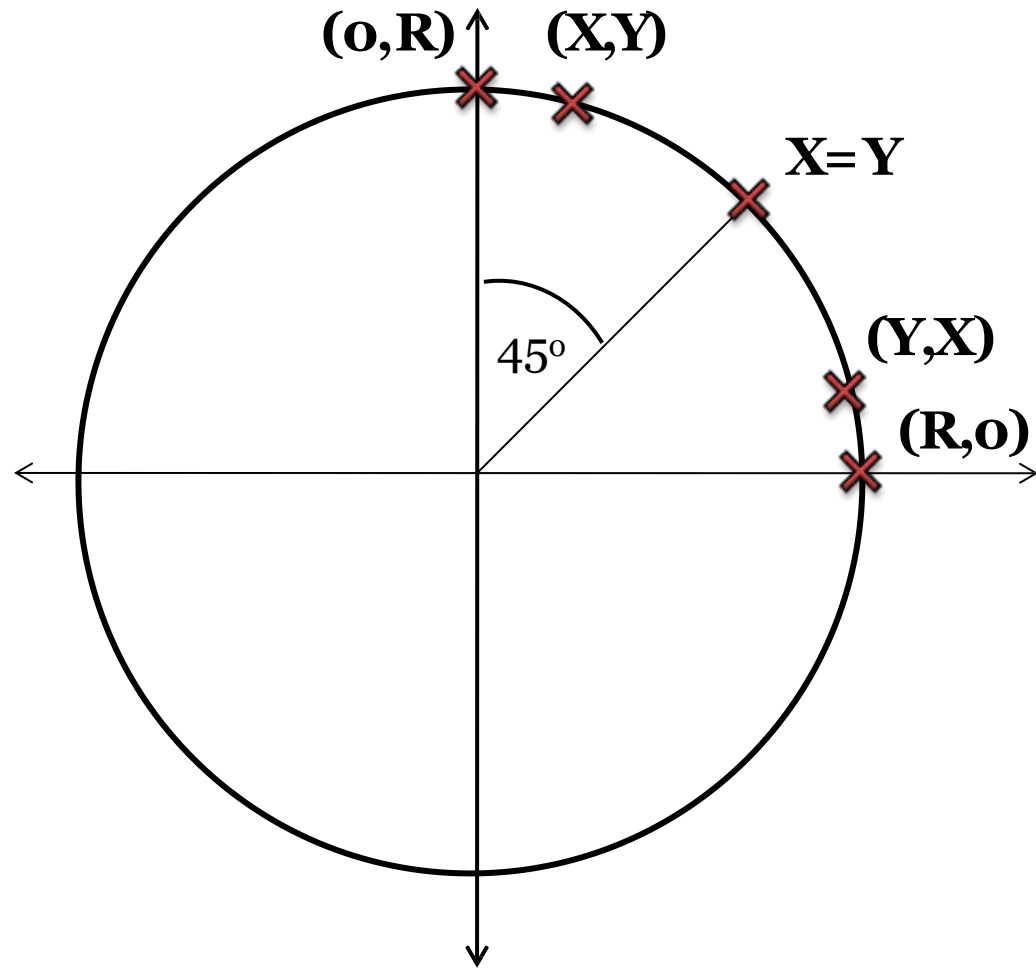
Observation



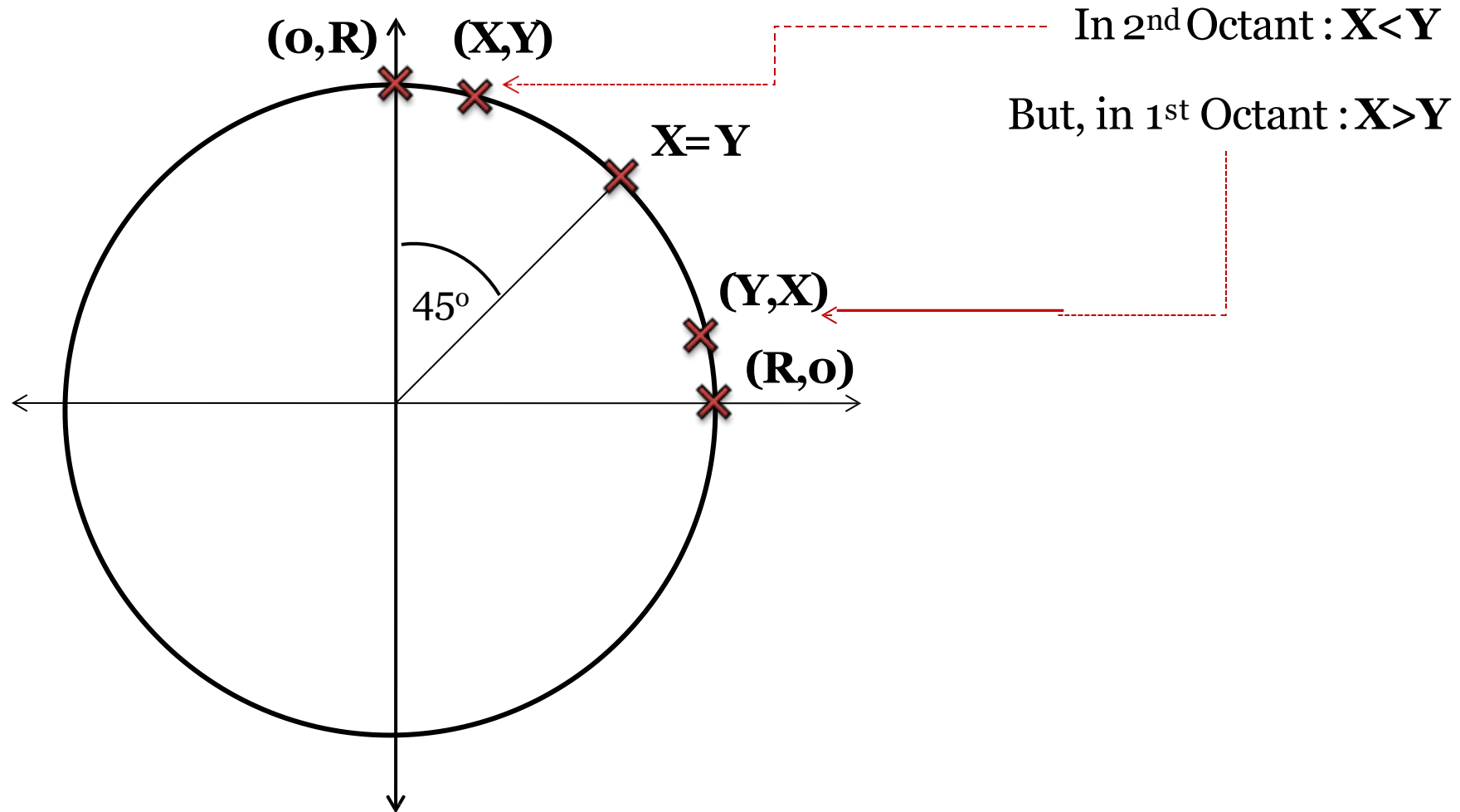
Observation

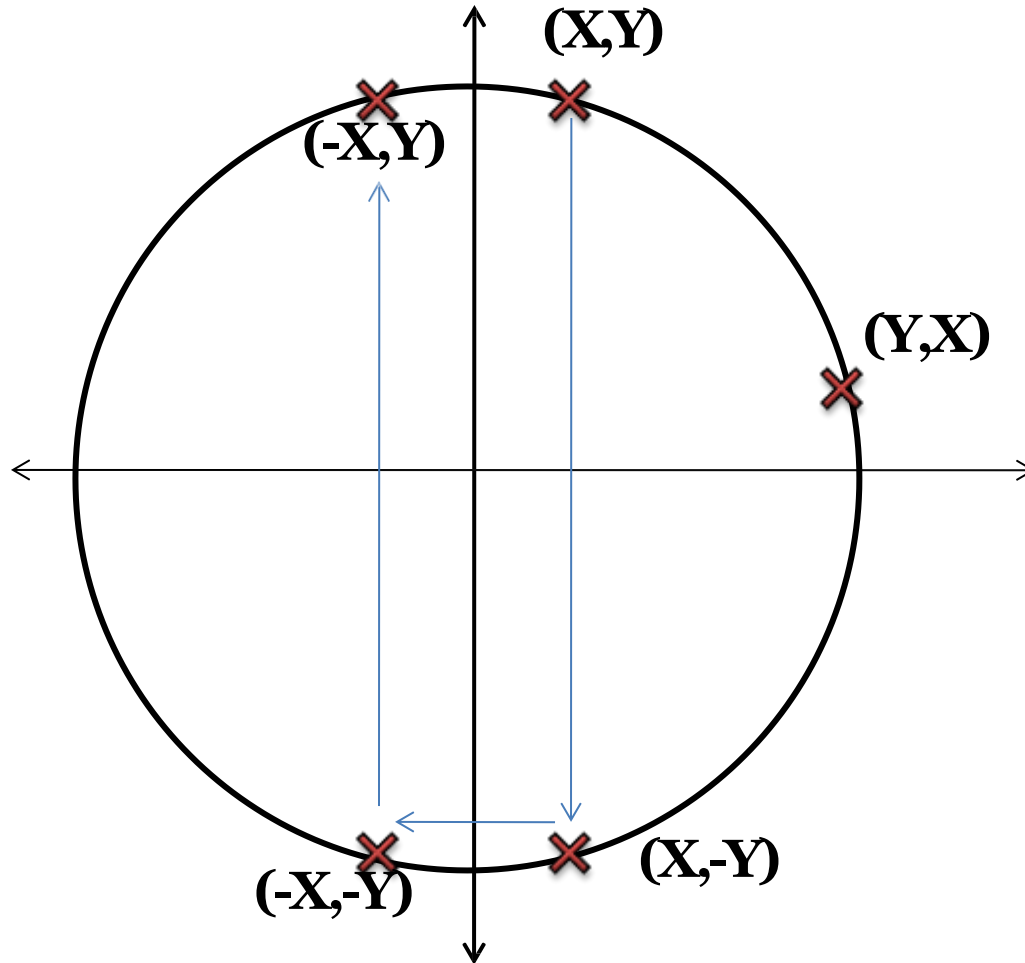


Observation



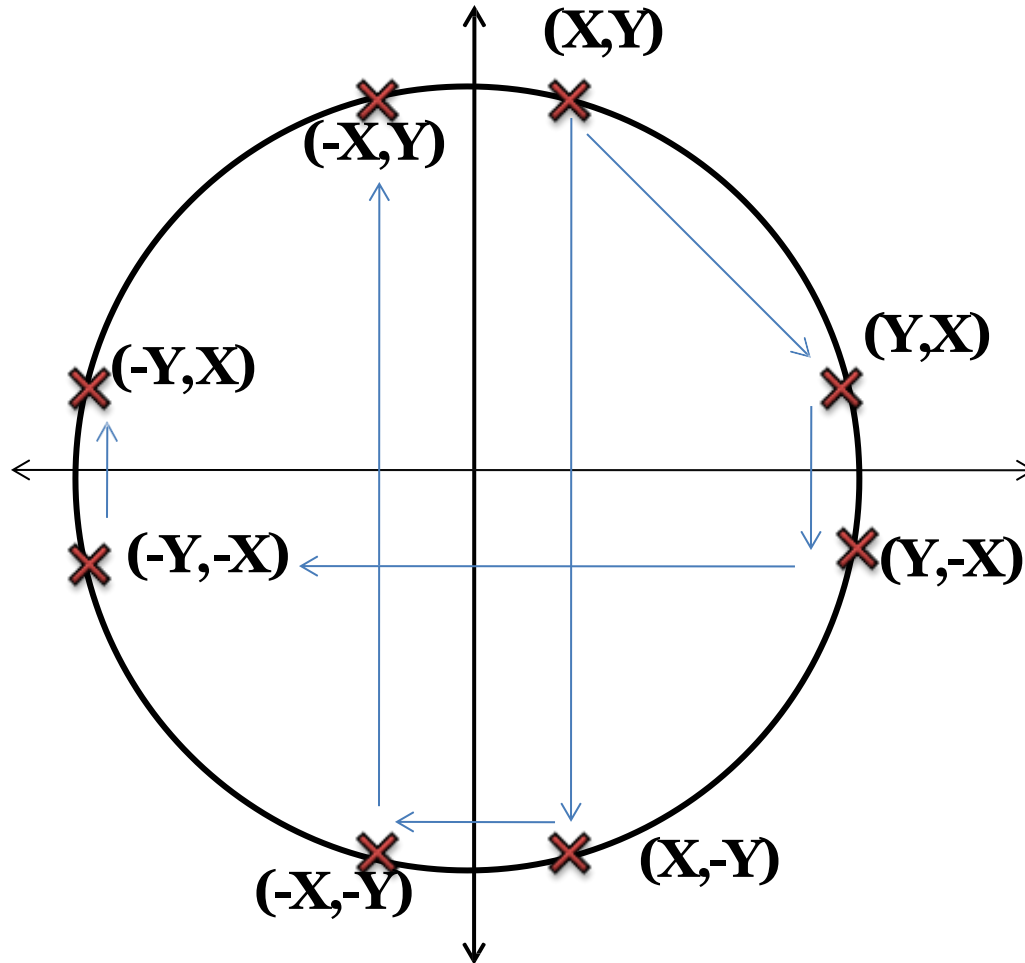
Observation





So, if we can obtain (X, Y) in 2nd octant, we can calculate the points-

- 7th Octant : $(X, -Y)$
- 6th Octant : $(-X, -Y)$
- 3rd Octant : $(-X, Y)$

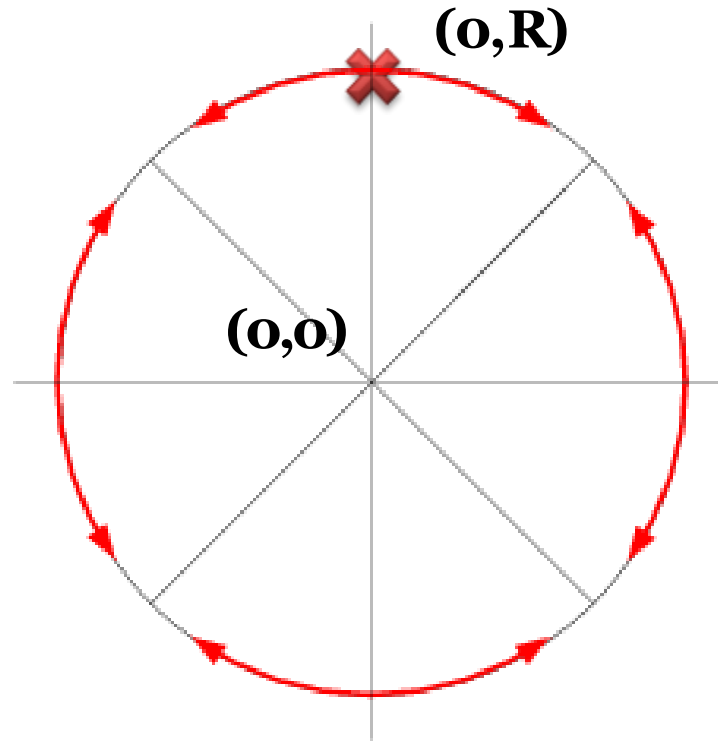


So, if we can obtain (X,Y) in 2nd octant, we can simply swap X and Y to get the points-

- 1st Octant : (Y,X)
- 8th Octant : $(Y,-X)$
- 5th Octant : $(-Y,-X)$
- 4th Octant : $(-Y,X)$

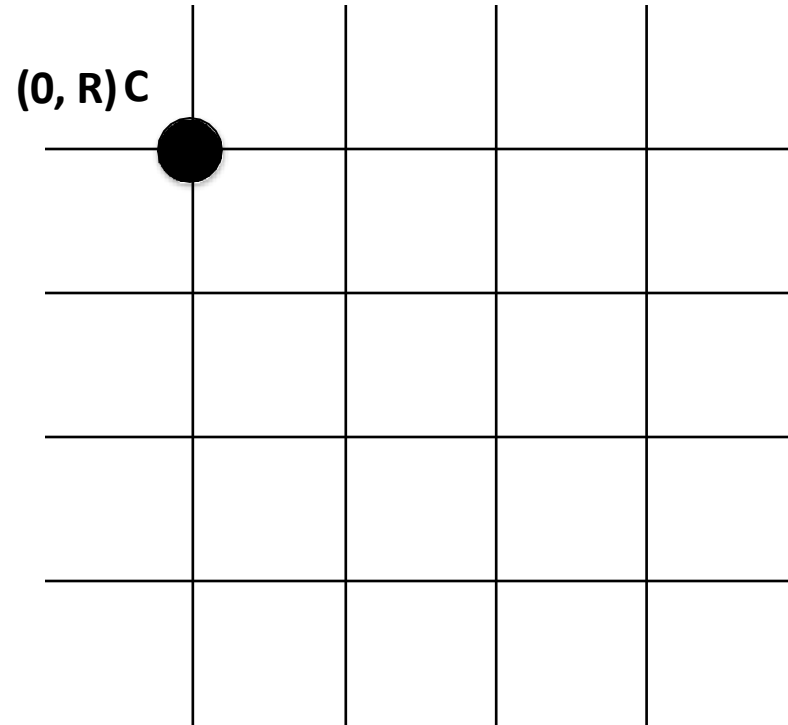
Drawing in all the octants

So, if we can obtain (X,Y) in 2nd octant, we can calculate the points in other 7 octants

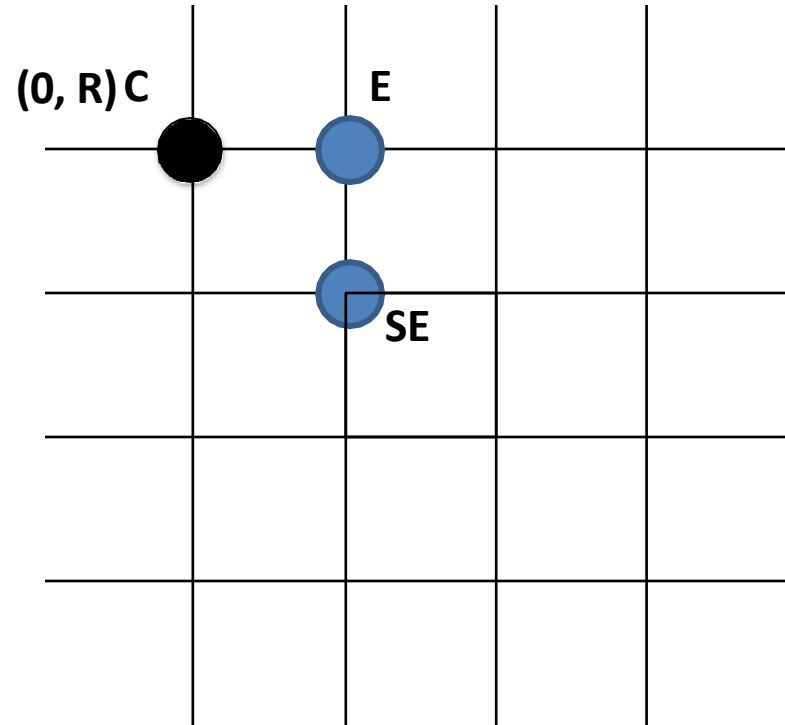


So, our target is to get the pixels of only 2nd octant of the circumference

Bresenham's Circle Drawing Algorithm: How it works

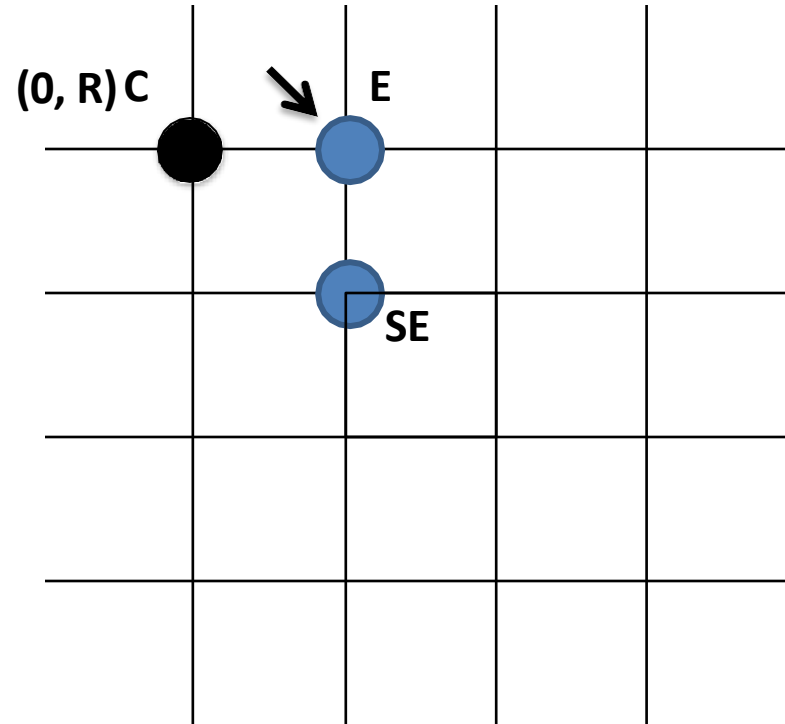


Bresenham's Circle Drawing Algorithm: How it works



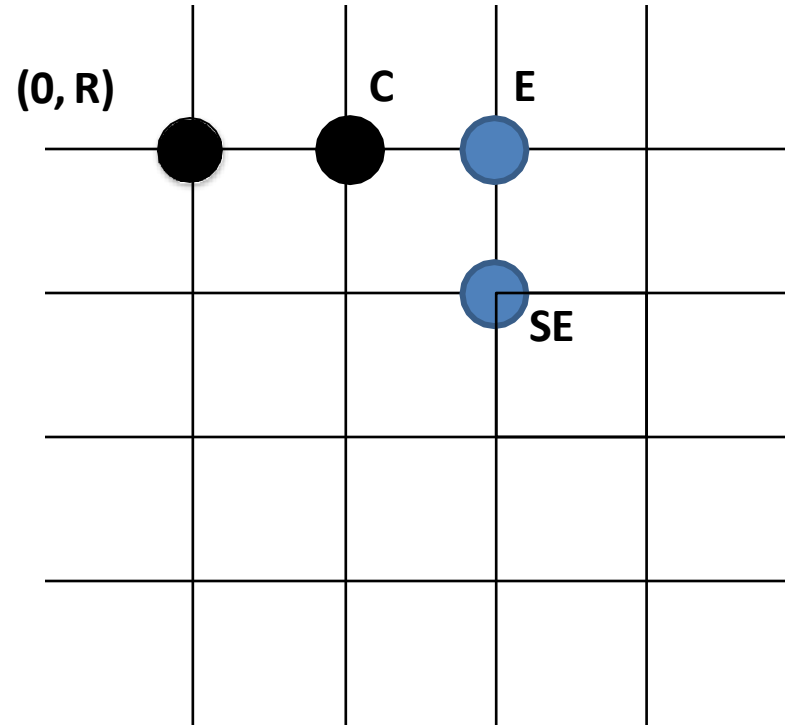
Next pixel is chosen
(from E or SE) to build
the line successively

Bresenham's Circle Drawing Algorithm: How it works



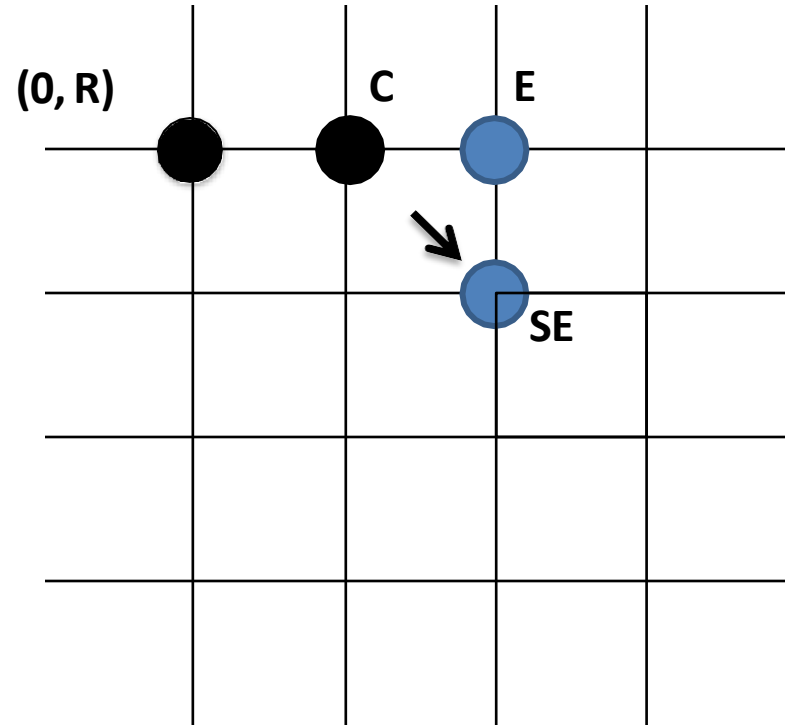
Next pixel is chosen
(from E or SE) to build
the line successively

Bresenham's Circle Drawing Algorithm: How it works



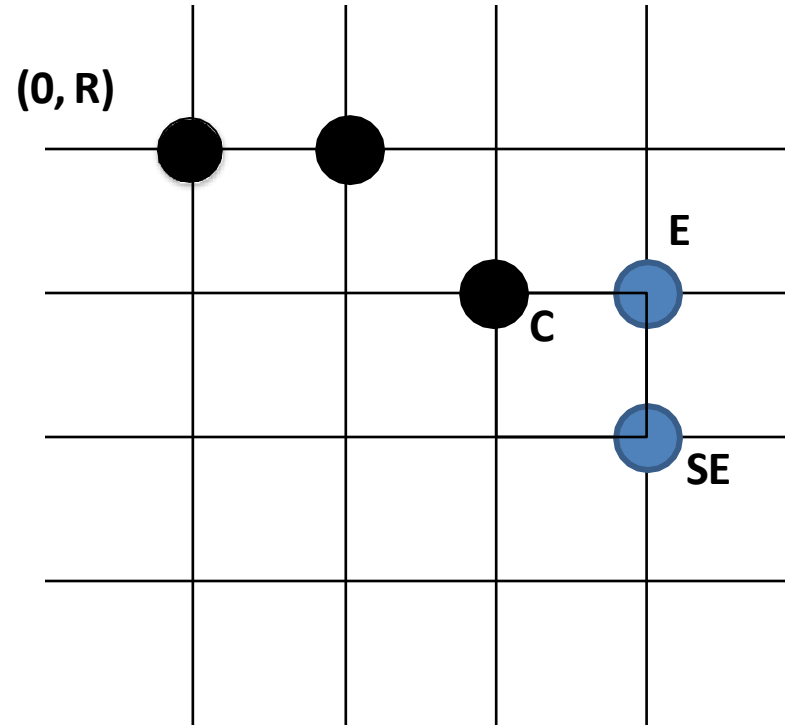
Next pixel is chosen
(from E or SE) to build
the line successively

Bresenham's Circle Drawing Algorithm: How it works



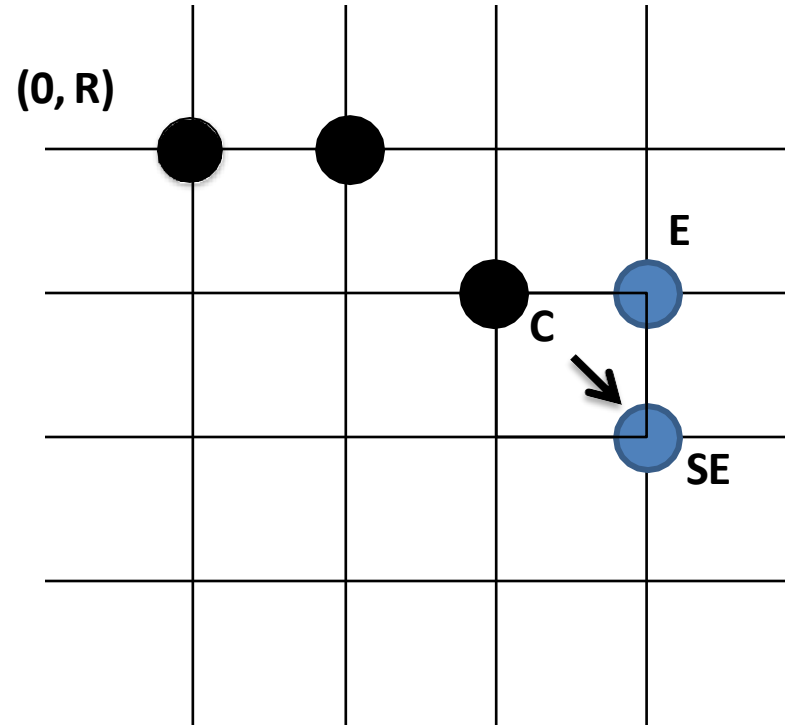
Next pixel is chosen
(from E or SE) to build
the line successively

Bresenham's Circle Drawing Algorithm: How it works



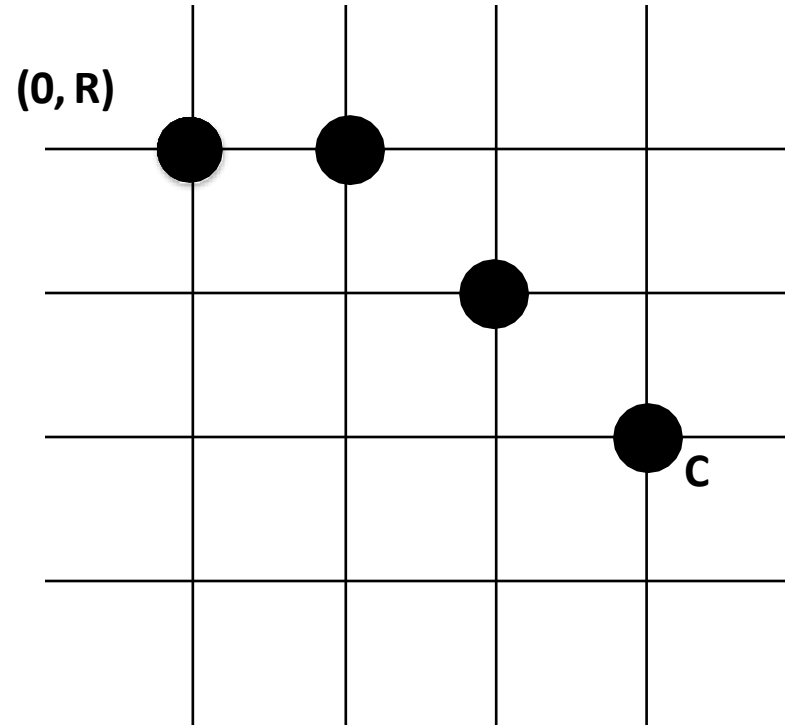
Next pixel is chosen
(from E or SE) to build
the line successively

Bresenham's Circle Drawing Algorithm: How it works



Next pixel is chosen
(from E or SE) to build
the line successively

Bresenham's Circle Drawing Algorithm: How it works



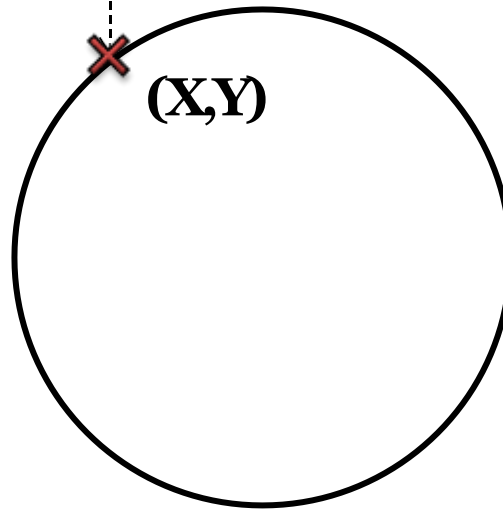
As we know that,
In 2nd Octant : $X < Y$
In 1st Octant : $X > Y$

**We will stop when $X > Y$,
that means when 2nd octant
is completed**

Equation of Circle and its function representation

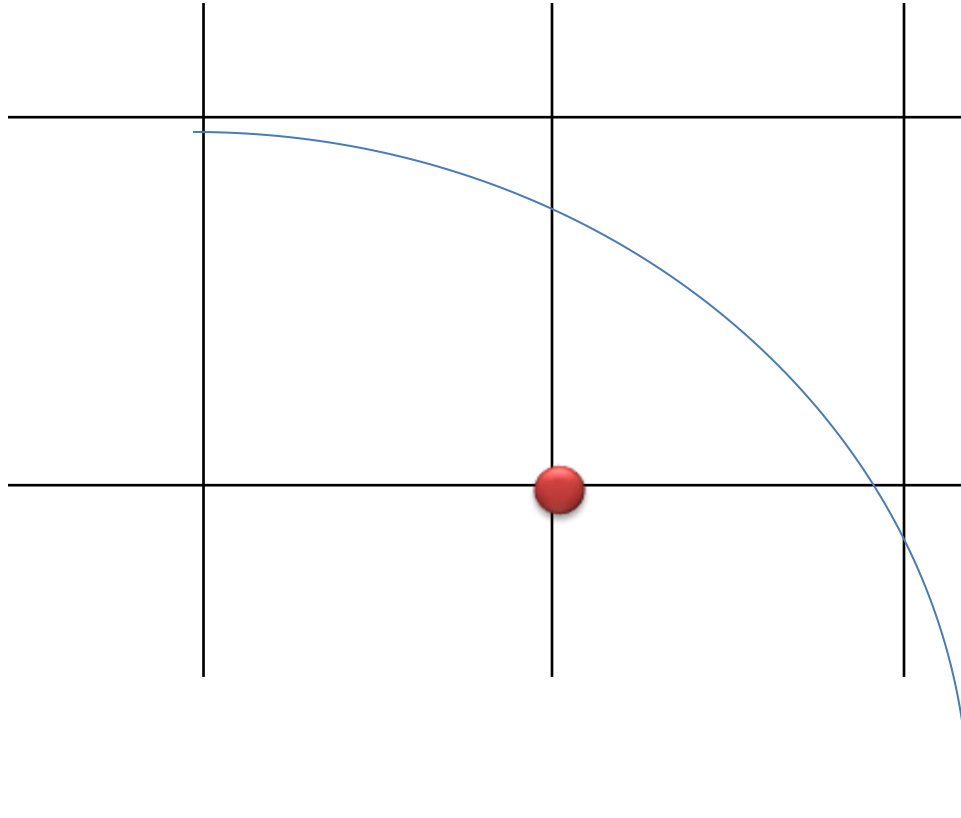
$$x^2 + y^2 = R^2$$

$$F(x, y) = x^2 + y^2 - R^2 = 0$$



Equation of Circle and its function representation

$$F(x, y) = x^2 + y^2 - R^2$$

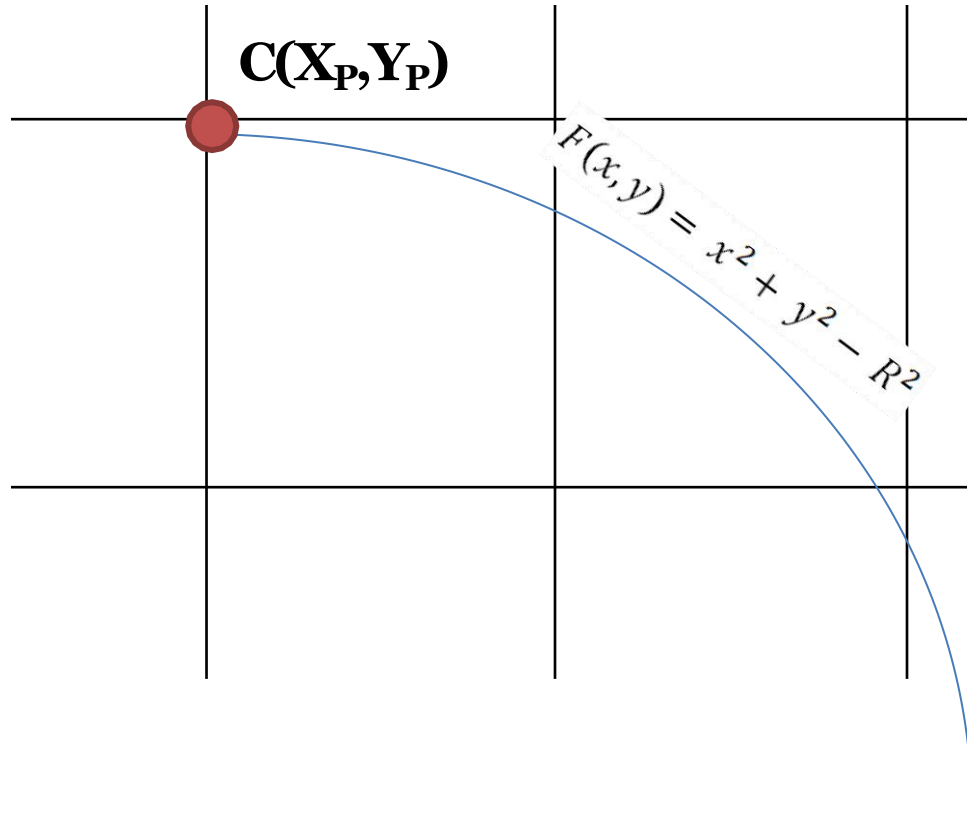


If $F(X, Y) = 0$, the point (X, Y) on the circle

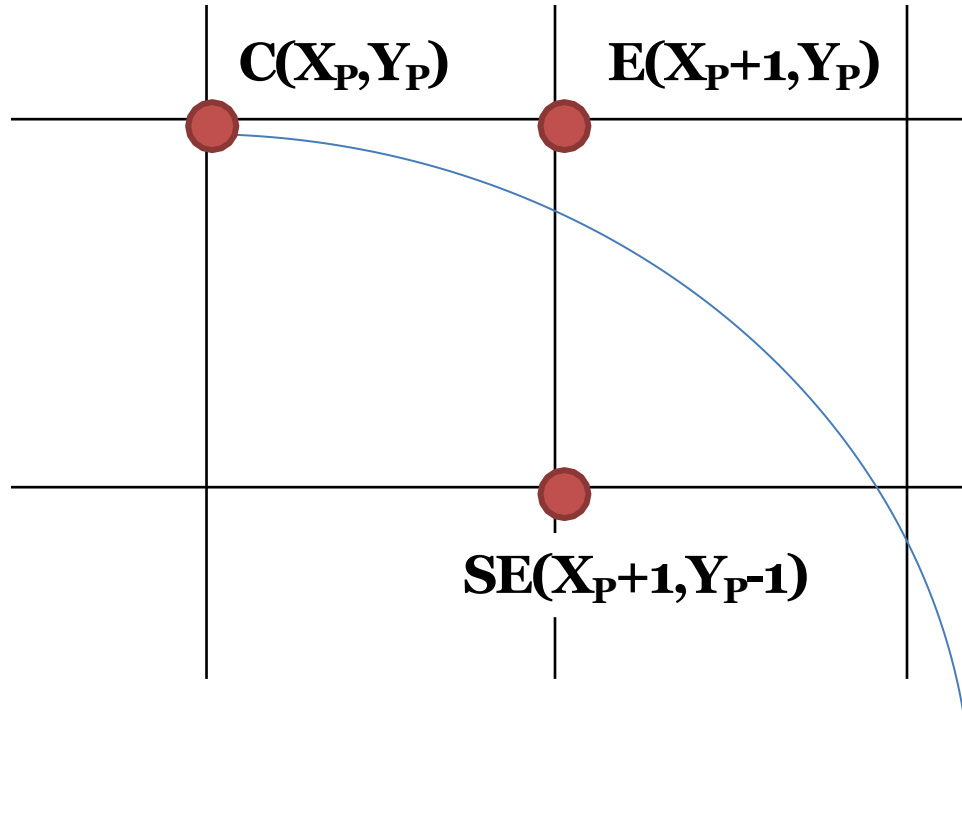
If $F(X, Y) > 0$, the point (X, Y) is outside the circle

If $F(X, Y) < 0$, the point (X, Y) is inside the circle

Selecting E or SE



Selecting E or SE

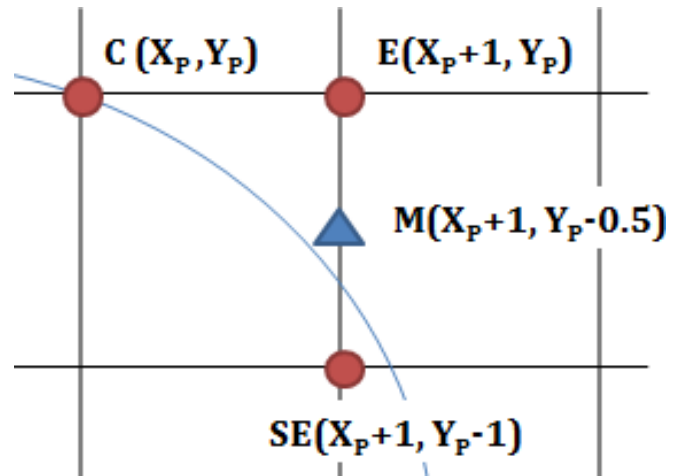


Selecting E or SE depends
on closeness to the
circumference.

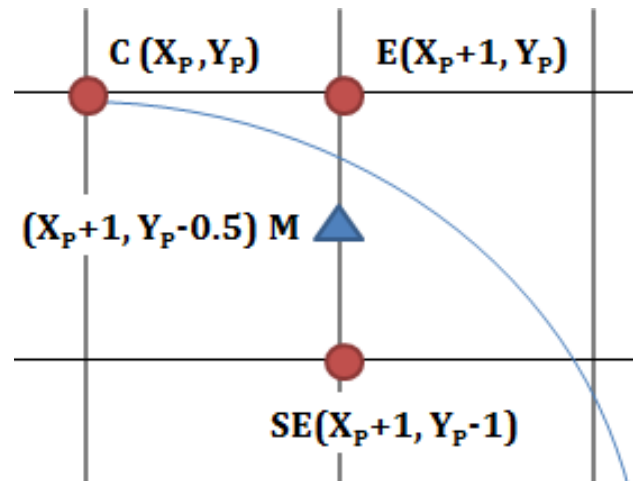
If E is closer to
circumference,
then E is selected

If SE is closer,
then SE is selected

Selecting E or SE using Mid Point Criteria



If midpoint M is outside the circle, SE is closer to the circumference,
So, **SE** is selected



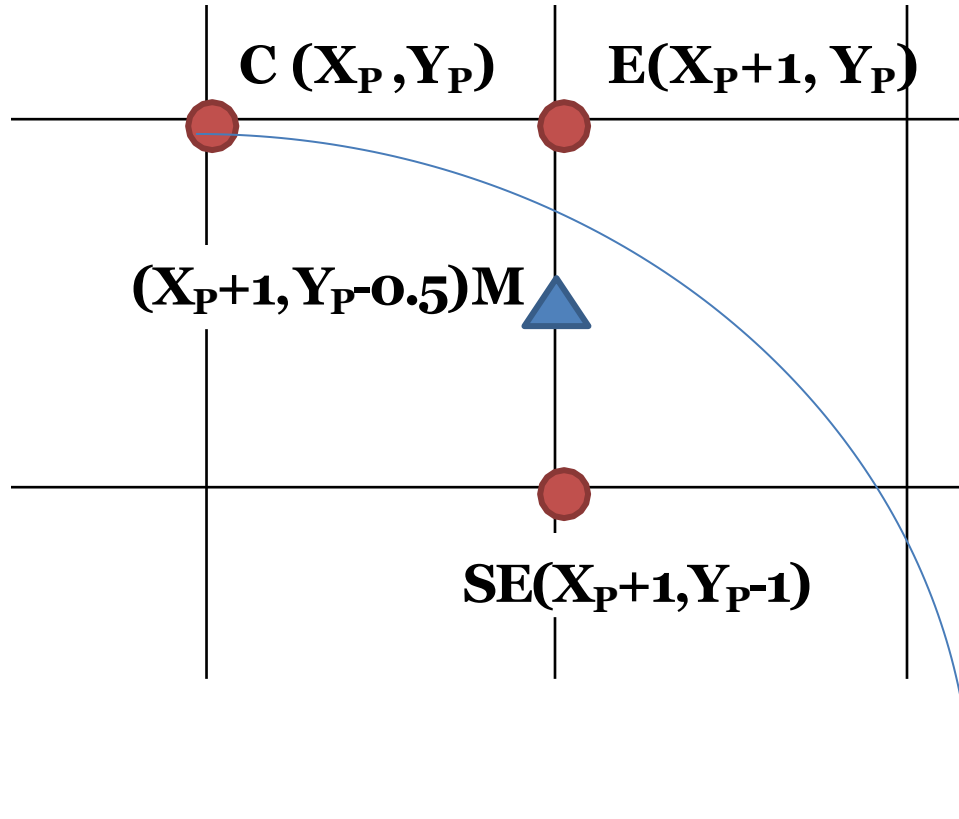
If midpoint M is inside the circle, E is closer to the circumference,
So, **E** is selected

Selecting E or SE using Mid Point Criteria

We know, $F(x, y) = x^2 + y^2 - R^2$

Lets put the mid point **M**'s coordinate in function $F(X, Y)$

$$F(M) = F(X_P + 1, Y_P - 0.5) = (X_P + 1)^2 + (Y_P - 0.5)^2 - R^2$$

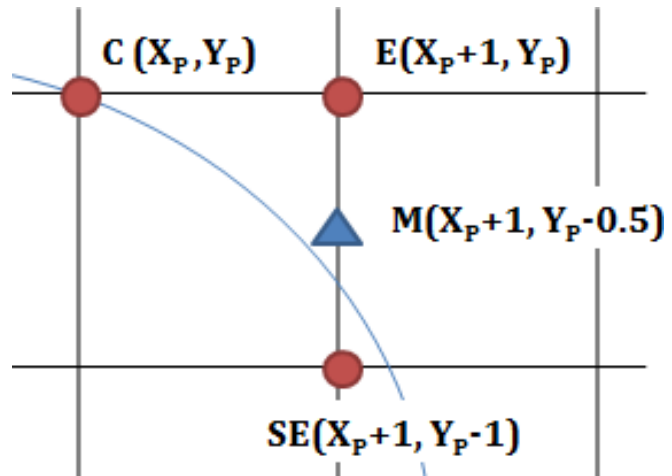


Lets store **F(M)** in a variable **d**

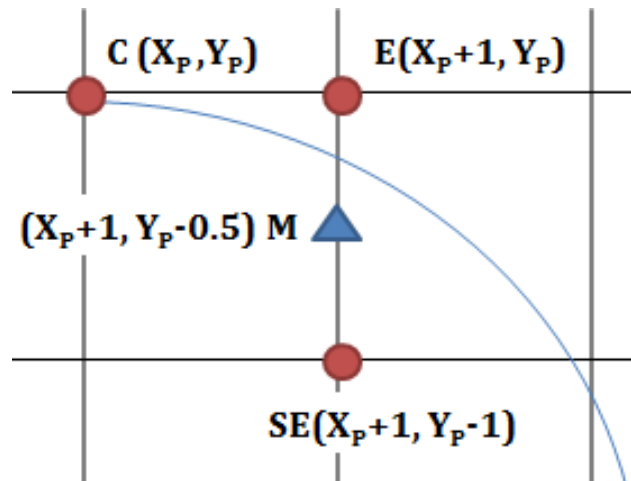
So, **d = F(M)**

d is called 'decision variable'

Selecting E or SE using Mid Point Criteria

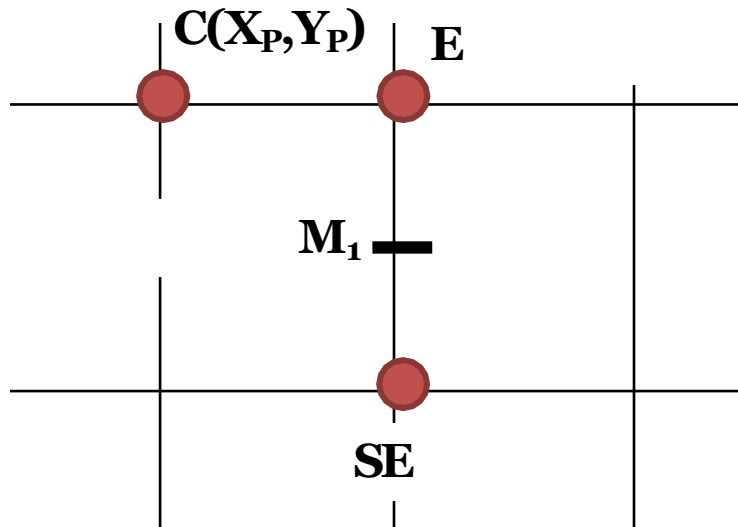


If $d \geq 0$, then midpoint M is outside the circle, SE is closer to the circumference,
So, **SE** is selected



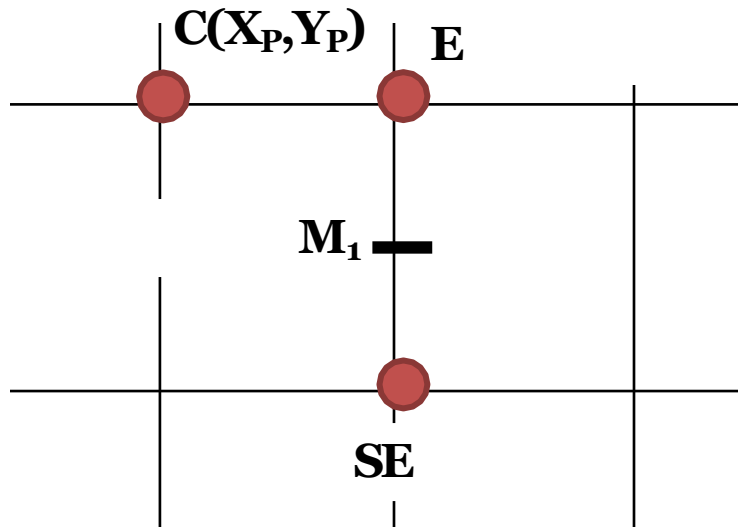
If $d < 0$, then midpoint M is inside the circle, E is closer to the circumference,
So, **E** is selected

Bresenham's Mid Point Criteria : Successive Updating (for selecting E)



$$\begin{aligned}d_1 &= F(M_1) \\&= F(X_P+1, Y_P-0.5) \\&= (X_P+1)^2 + (Y_P-0.5)^2 - R^2\end{aligned}$$

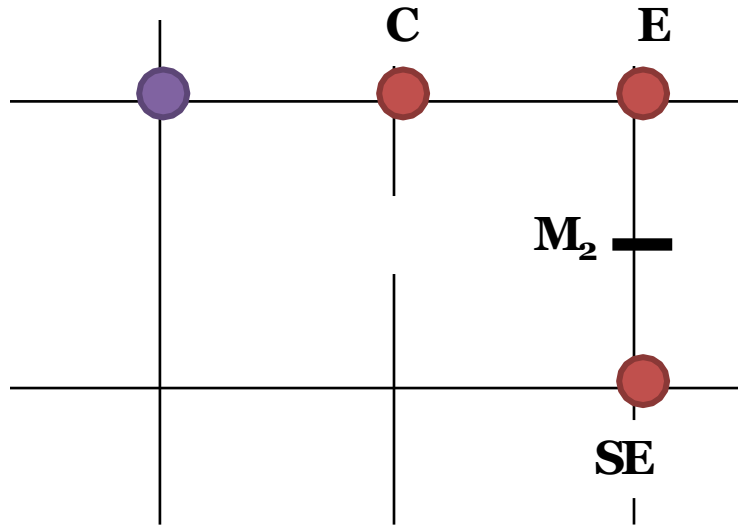
Bresenham's Mid Point Criteria : Successive Updating (for selecting E)



$$\begin{aligned}d_1 &= F(M_1) \\&= F(X_P+1, Y_P-0.5) \\&= (X_P+1)^2 + (Y_P-0.5)^2 - R^2\end{aligned}$$

If $d_1 < 0$, $E(X_P+1, Y_P)$

Bresenham's Mid Point Criteria : Successive Updating (for selecting E)

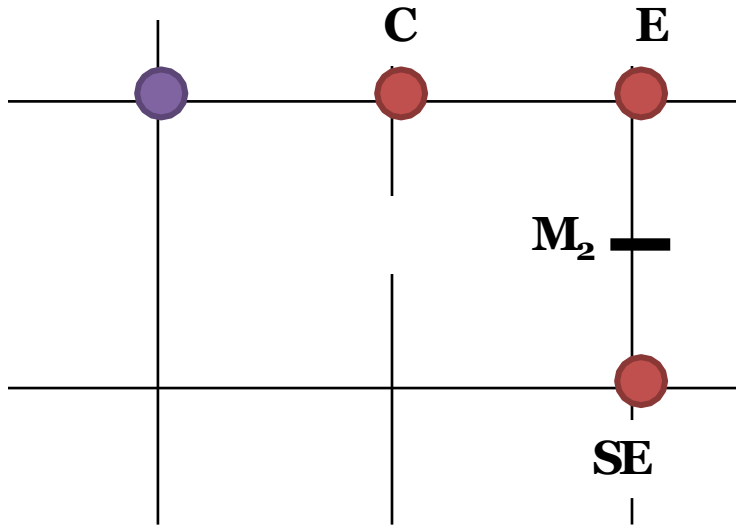


$$\begin{aligned} d_1 &= F(M_1) \\ &= F(X_P+1, Y_P-0.5) \\ &= (X_P+1)^2 + (Y_P-0.5)^2 - R^2 \end{aligned}$$

If $d_1 < 0$, $E(X_P+1, Y_P)$

$$\begin{aligned} d_2 &= F(M_2) \\ &= F(X_P+2, Y_P-0.5) \\ &= (X_P+2)^2 + (Y_P-0.5)^2 - R^2 \\ &= X_P^2 + 4X_P + 4 + (Y_P-0.5)^2 - R^2 \\ &= X_P^2 + 2X_P + 1 + (Y_P-0.5)^2 - R^2 + 2X_P + 3 \\ &= d_1 + (2X_P + 3) \end{aligned}$$

Bresenham's Mid Point Criteria : Successive Updating (for selecting E)



$$\begin{aligned} d_1 &= F(M_1) \\ &= F(X_p+1, Y_p-0.5) \\ &= (X_p+1)^2 + (Y_p-0.5)^2 - R^2 \end{aligned}$$

If $d_1 < 0$, $E(X_p+1, Y_p)$

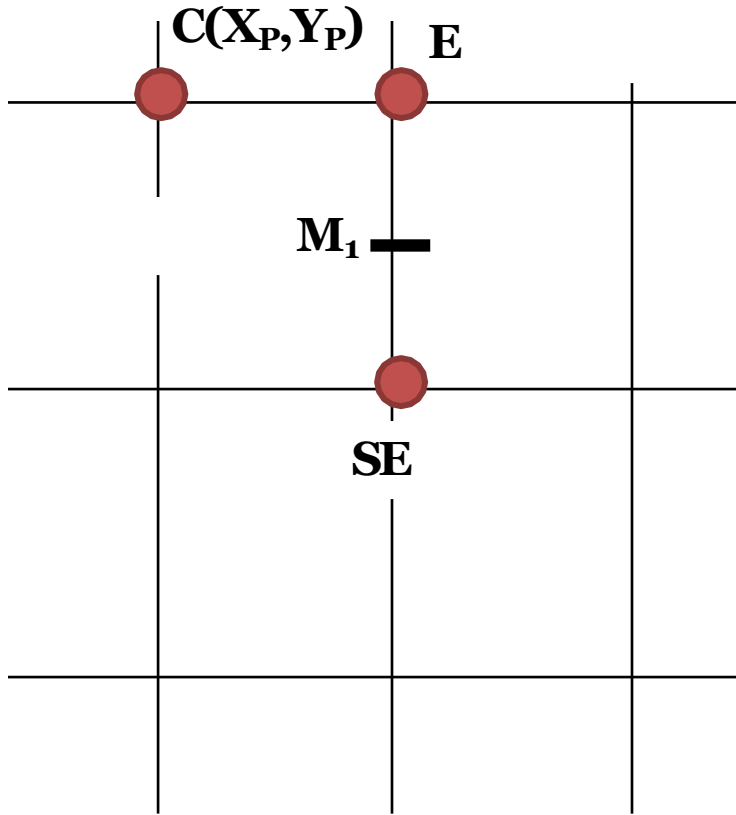
$$\begin{aligned} d_2 &= F(M_2) \\ &= F(X_p+2, Y_p-0.5) \\ &= (X_p+2)^2 + (Y_p-0.5)^2 - R^2 \\ &= X_p^2 + 4X_p + 4 + (Y_p-0.5)^2 - R^2 \\ &= X_p^2 + 2X_p + 1 + (Y_p-0.5)^2 - R^2 + 2X_p + 3 \\ &= d_1 + (2X_p + 3) \end{aligned}$$

Every iteration after **selecting E**, we can successively update our decision variable with-

$$\mathbf{d_{NEW} = d_{OLD} + (2X_P + 3)}$$

Bresenham's Mid Point Criteria : Successive Updating (for selecting SE)

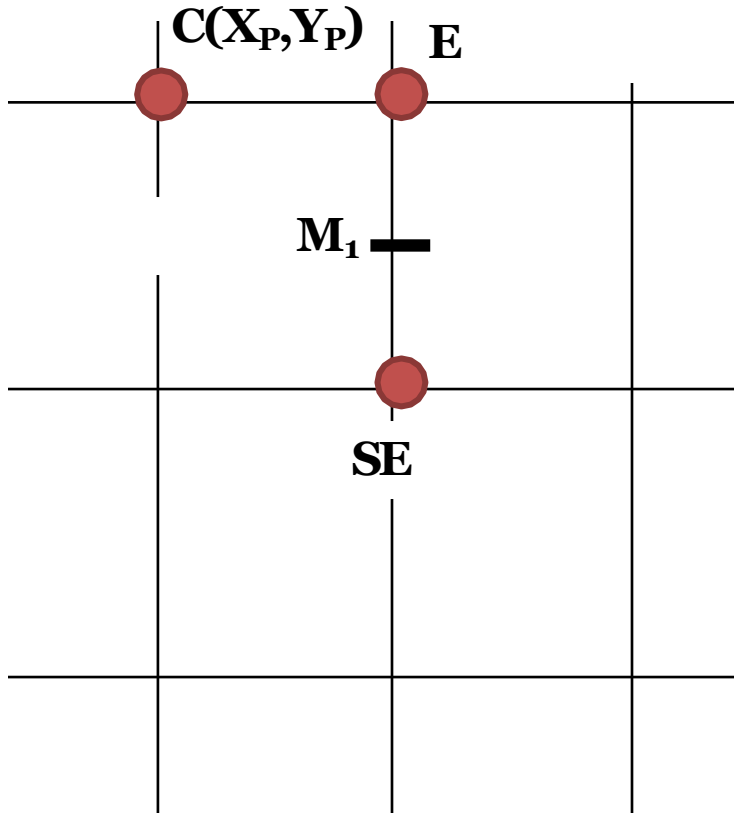
$$\begin{aligned}d_1 &= F(M_1) \\&= F(X_P+1, Y_P-0.5) \\&= (X_P+1)^2 + (Y_P-0.5)^2 - R^2\end{aligned}$$



Bresenham's Mid Point Criteria : Successive Updating (for selecting SE)

$$\begin{aligned}d_1 &= F(M_1) \\&= F(X_P+1, Y_P-0.5) \\&= (X_P+1)^2 + (Y_P-0.5)^2 - R^2\end{aligned}$$

If $d_1 \geq 0$, SE($X_P=X_P+1, Y_P-1$)



Bresenham's Mid Point Criteria : Successive Updating (for selecting SE)

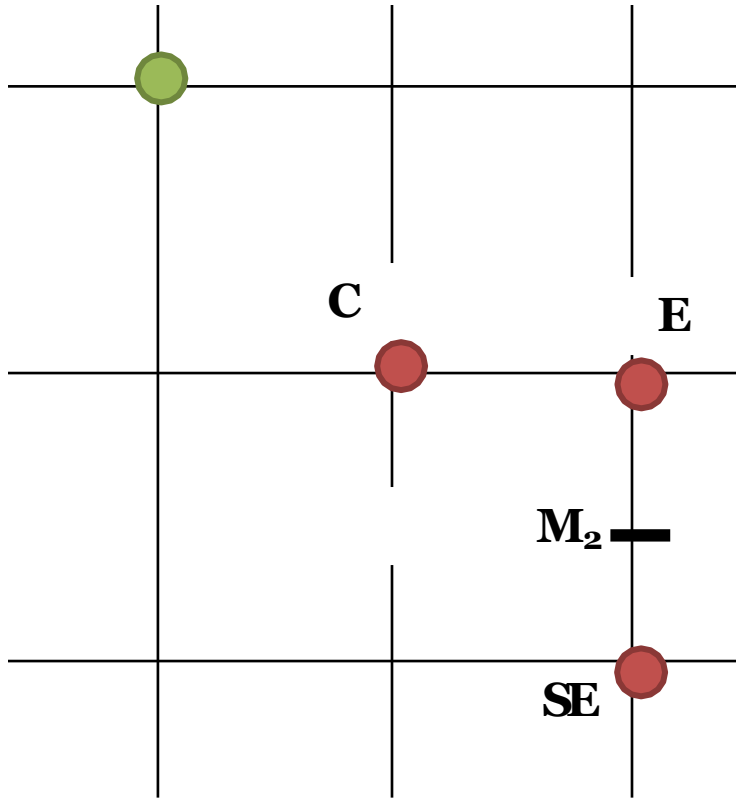
$$\begin{aligned}d_1 &= F(M_1) \\&= F(X_P+1, Y_P-0.5) \\&= (X_P+1)^2 + (Y_P-0.5)^2 - R^2\end{aligned}$$

If $d_1 \geq 0$, SE($X_P=X_P+1, Y_P-1$)

$$d_2 = F(M_2)$$

.... DIY....

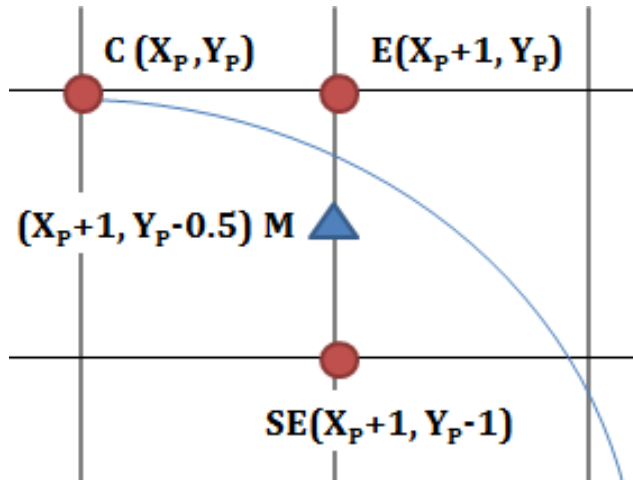
$$= d_1 + (2X_P - 2Y_P + 5)$$



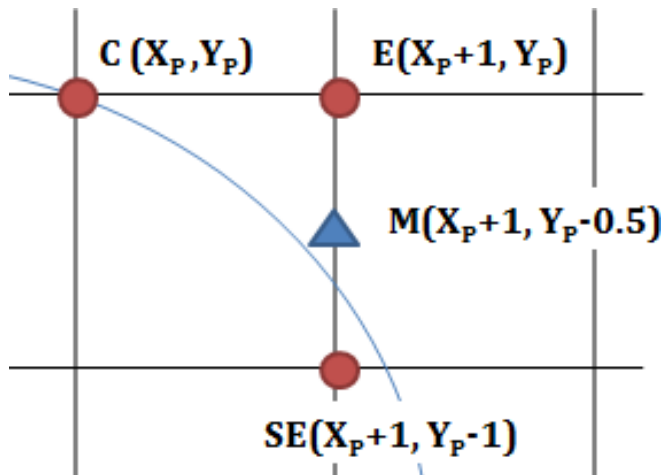
Every iteration after **selecting NE**, we can successively update our decision variable with-

$$d_{\text{NEW}} = d_{\text{OLD}} + (2X_P - 2Y_P + 5)$$

Bresenham's Mid Point Criteria : Successive Updating (summary)

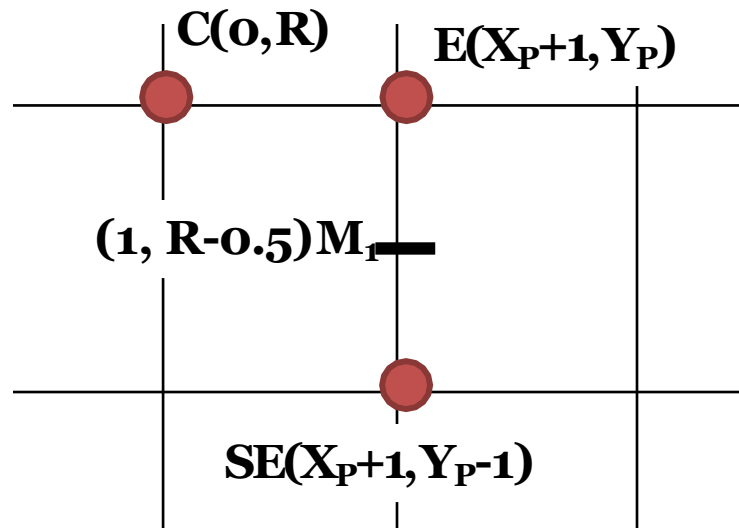


If $d < 0$, then midpoint M is inside the circle, E is closer to the circumference,
So, **E** is selected and do-
 $d = d + \Delta E$
Where, $\Delta E = 2X_P + 3$



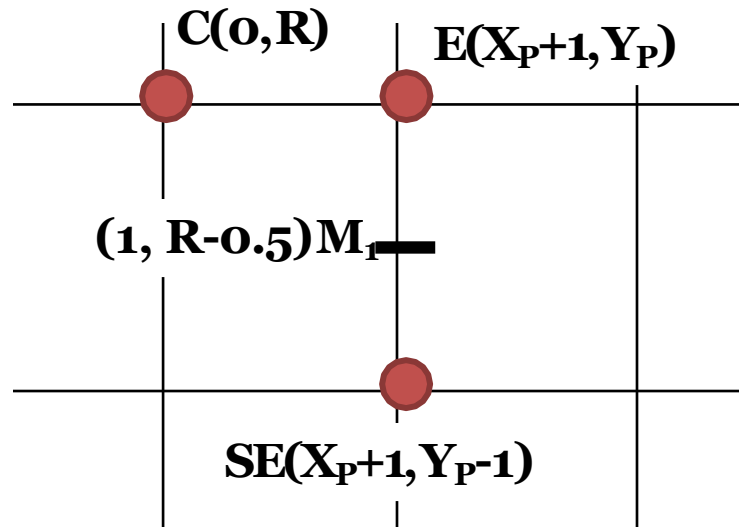
If $d \geq 0$, then midpoint M is outside the circle, SE is closer to the circumference,
So, **SE** is selected and do-
 $d = d + \Delta SE$
Where, $\Delta SE = 2X_P - 2Y_P + 5$

Initialization



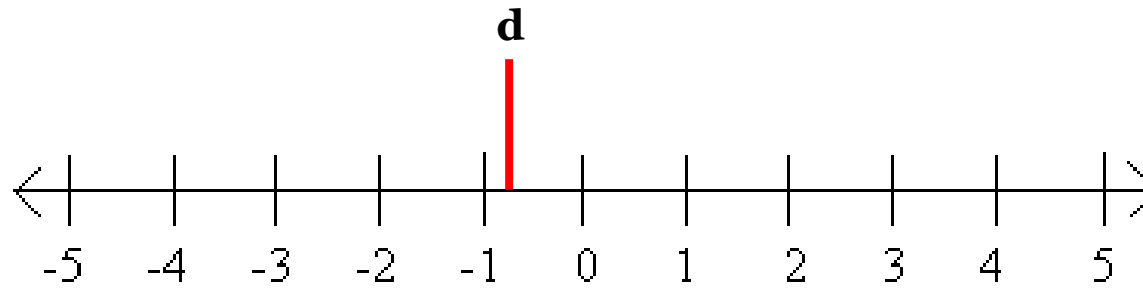
$$\begin{aligned}d_{\text{INIT}} &= F(M_1) \\&= F(1, R-0.5) \\&= (1)^2 + (R-0.5)^2 - R^2 \\&= 1 + R^2 - R + 0.25 - R^2 \\&= 1.25 - R\end{aligned}$$

Initialization



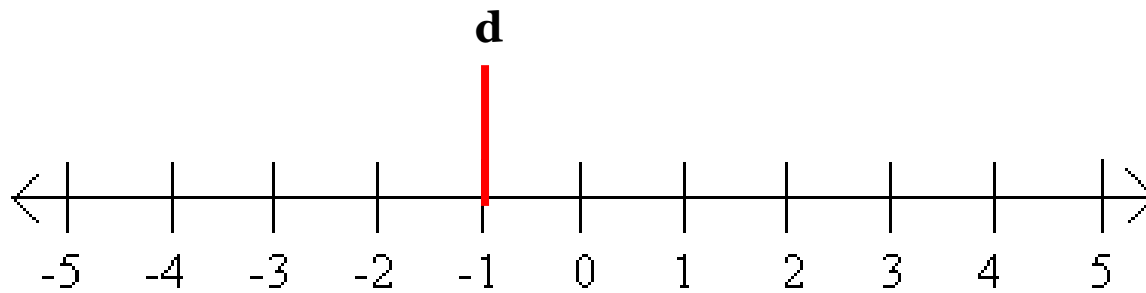
$$\begin{aligned}d_{\text{INIT}} &= F(M_1) \\&= F(1, R-0.5) \\&= (1)^2 + (R-0.5)^2 - R^2 \\&= 1 + R^2 - R + 0.25 - R^2 \\&= \mathbf{1.25 - R} \\&\approx \mathbf{1 - R}\end{aligned}$$

Initialization



$$R = 2$$

$$d = 1.25 - R = -0.75$$



$$R = 2$$

$$d = 1 - R = -1$$

So, finally.....

$$\mathbf{d}_{\text{INIT}} = \mathbf{1} - \mathbf{R}$$

If $\mathbf{d} < \mathbf{o}$, then $\mathbf{E}$ is selected, $\mathbf{d} = \mathbf{d} + \Delta\mathbf{E}$

If $\mathbf{d} \geq \mathbf{o}$, then $\mathbf{SE}$ is selected, $\mathbf{d} = \mathbf{d} + \Delta\mathbf{SE}$

Where,

$$\Delta\mathbf{E} = 2\mathbf{X}_p + 3$$

$$\Delta\mathbf{SE} = 2\mathbf{X}_p - 2\mathbf{Y}_p + 5$$

```
void MidpointCircle(int radius)
{
    int x = 0;
    int y = radius ;
    int d = 1 - radius ;
    CirclePoints(x, y);
    while (y > x)
    {
        if (d < 0) /* Select E */
            d = d + 2 * x + 3;
        else
        {
            /* Select SE */
            d = d + 2 * (x - y) + 5;
            y = y - 1;
        }
        x = x + 1;
        CirclePoints(x, y);
    }
}
```

```
void MidpointCircle(int radius)
{
    int x = 0;
    int y = radius ;
    int d = 1 - radius ;
    CirclePoints(x,y);
    while (y > x)
    {
        if (d < 0) /* Select E */
            d = d + 2 * x + 3;
        else
        {
            /* Select SE */
            d = d + 2 * ( x - y ) + 5;
            y = y - 1;
        }
        x = x + 1;
        CirclePoints(x,y);
    }
}
```

```
CirclePoints (x,y)
    Plotpoint(x,y) ;
    Plotpoint (x,-y);
    Plotpoint(-x,y) ;
    Plotpoint(-x, -y);
    Plotpoint(y,x) ;
    Plotpoint(y, -x);
    Plotpoint(-y, x);
    Plotpoint( -y, -x);
end
```

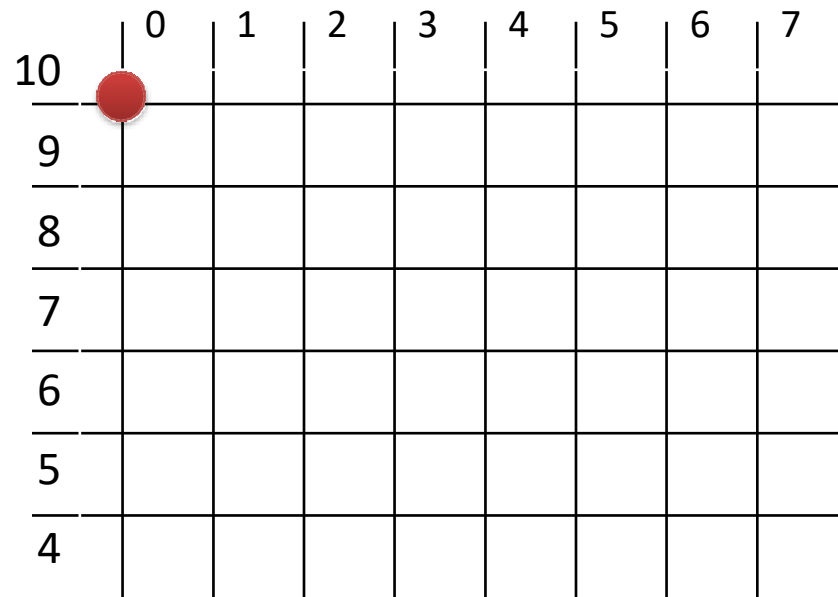
Example

	0	1	2	3	4	5	6	7
10								
9								
8								
7								
6								
5								
4								

Given:

Radius , $R = 10$

Example



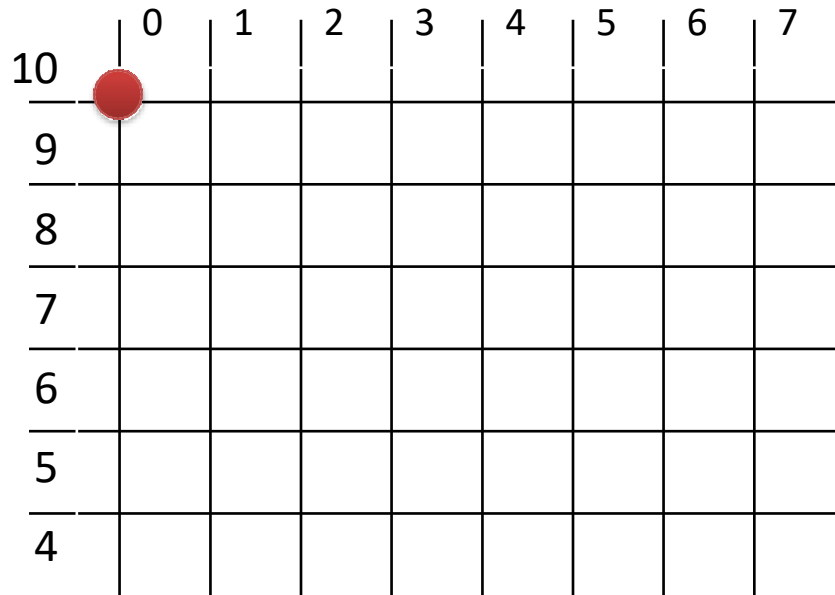
Given:

Radius, $R = 10$

$(x, y) = (0, 10)$

$h = 1 \quad -R = -9$

Example



Given:

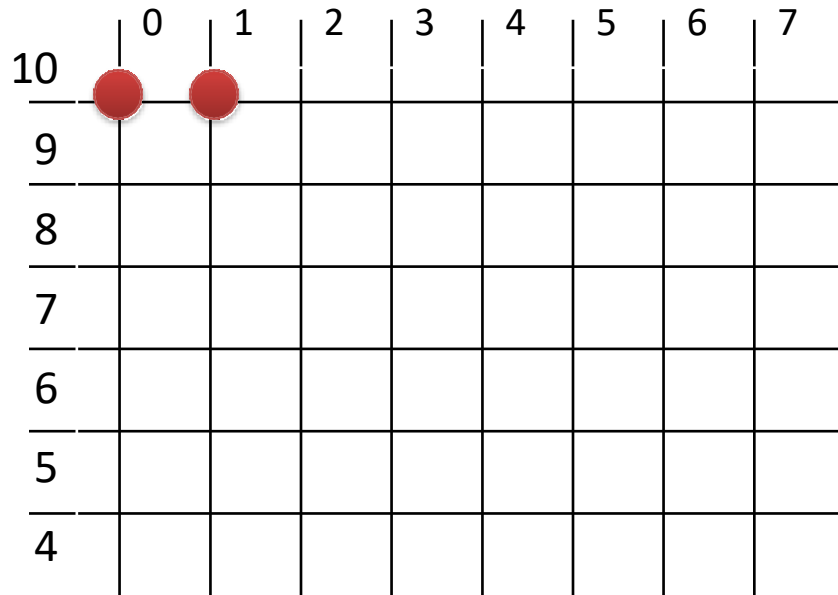
Radius , $R = 10$

$(x,y)=(0,10)$

$h = 1 - R = -9$

K	1						
2x	0						
2y	20						
h							
(x,y)							

Example



Given:

Radius , $R = 10$

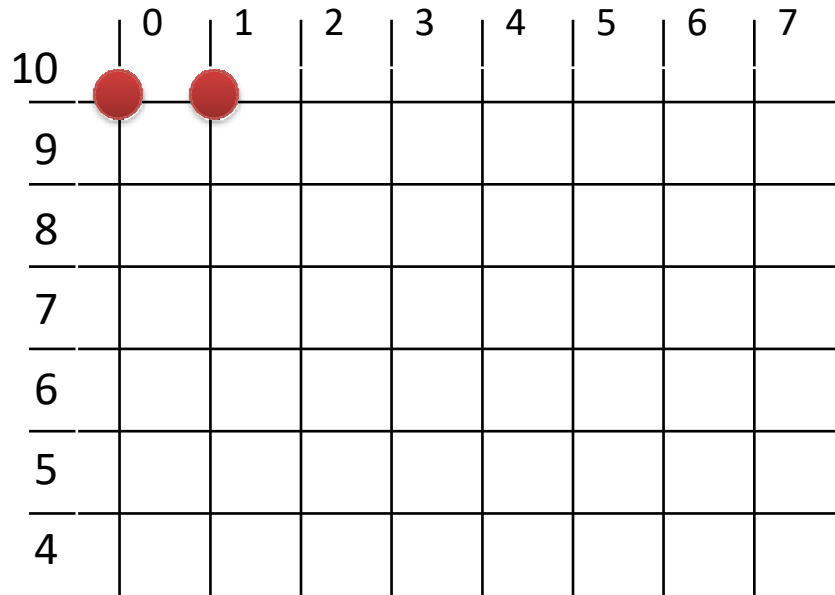
$(x,y)=(0,10)$

$h = 1 - R = -9$

K	1						
2x	0						
2y	20						
h							
(x,y)	E(1,10)						

$h \leq 0, E$

Example



Given:

Radius , $R = 10$

$(x,y)=(0,10)$

$h = 1 - R = -9$

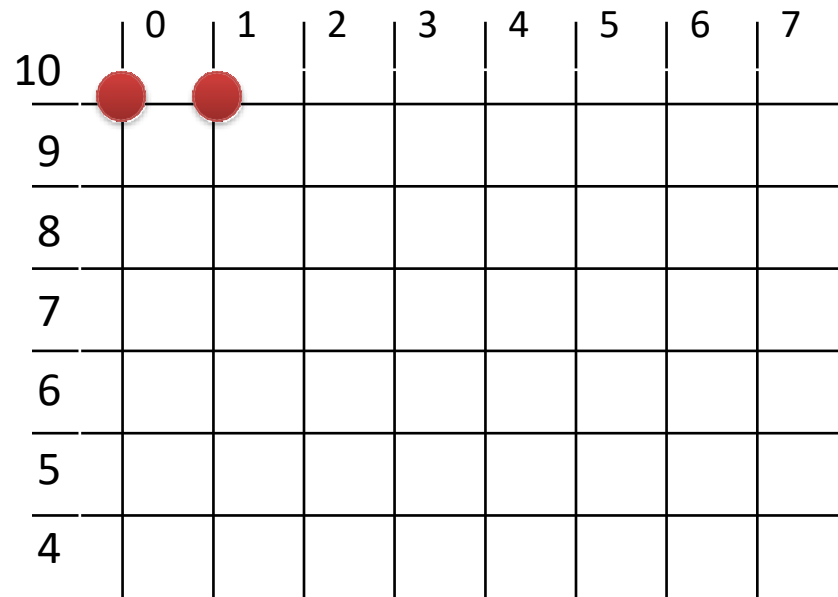
$h = h + \Delta E = h + 2x + 3$

$= -9 + 0 + 3$

$= -6$

K	1						
2x	0						
2y	20						
h	-6						
(x,y)	E(1,10)						

Example



Given:

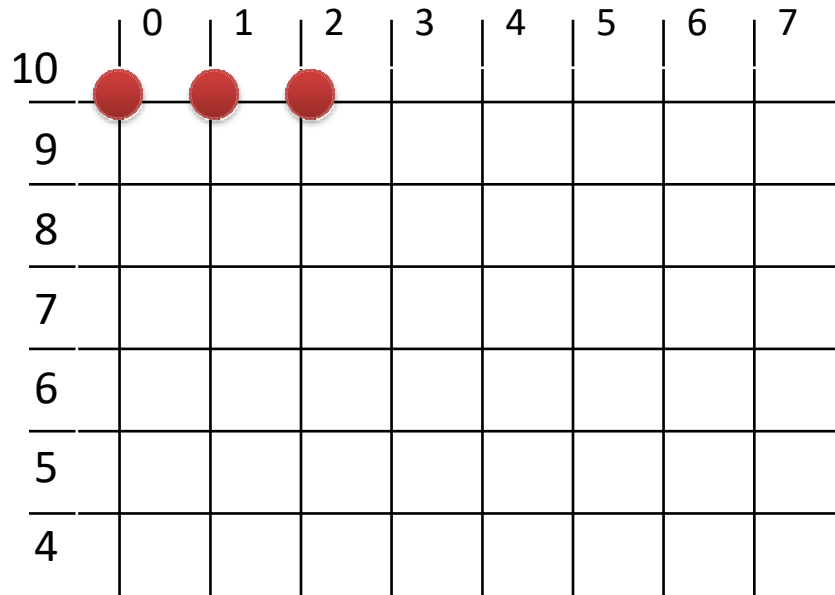
Radius , $R = 10$

$(x,y)=(0,10)$

$h = 1 - R = -9$

K	1	2					
2x	0	2					
2y	20	20					
h	-6						
(x,y)	E(1,10)						

Example



Given:

Radius, $R = 10$

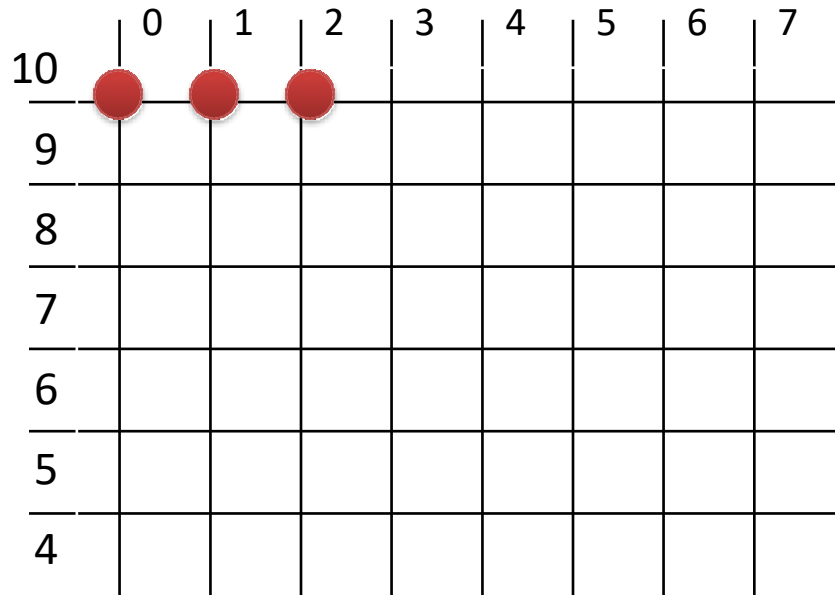
$(x, y) = (0, 10)$

$h = 1 - R = -9$

K	1	2					
2x	0	2					
2y	20	20					
h	↙ -6						
(x, y)	E(1, 10)	E(2, 10)					

$h \leq 0, E$

Example



Given:

Radius , $R = 10$

$(x,y)=(0,10)$

$h = 1 - R = -9$

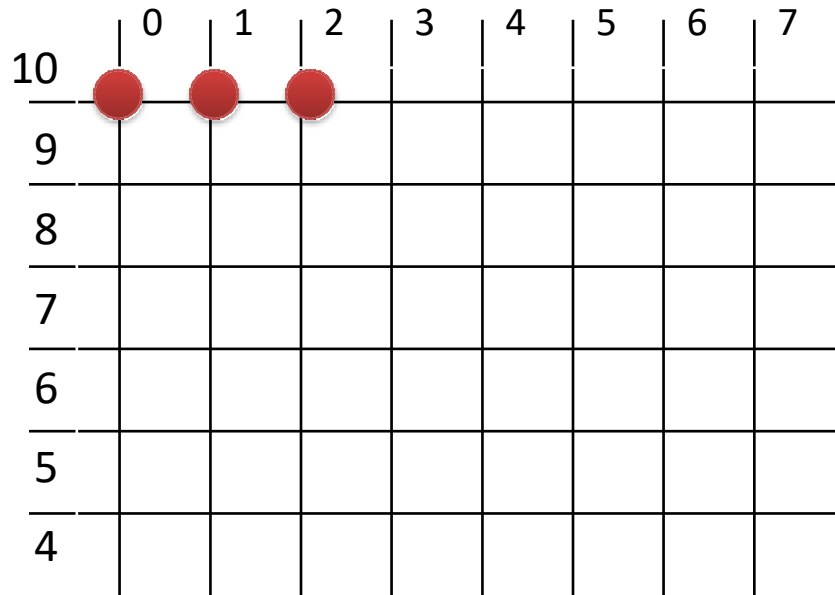
$h = h + \Delta E = h + 2x + 3$

$= -6 + 2 + 3$

$= -1$

K	1	2					
2x	0	2					
2y	20	20					
h	-6	-1					
(x,y)	E(1,10)	E(2,10)					

Example



Given:

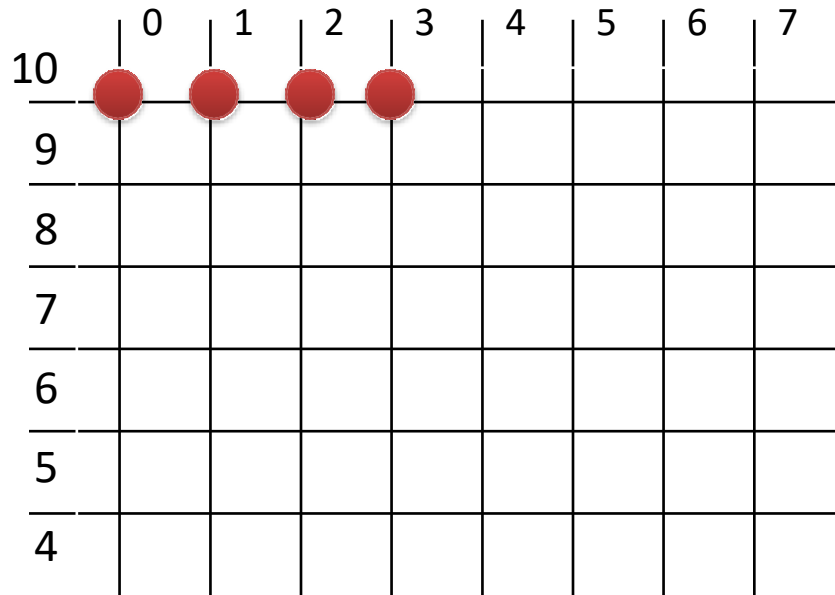
Radius , $R = 10$

$(x,y)=(0,10)$

$h = 1 - R = -9$

K	1	2	3				
2x	0	2	4				
2y	20	20	20				
h	-6	-1					
(x,y)	E(1,10)	E(2,10)					

Example



Given:

Radius, $R = 10$

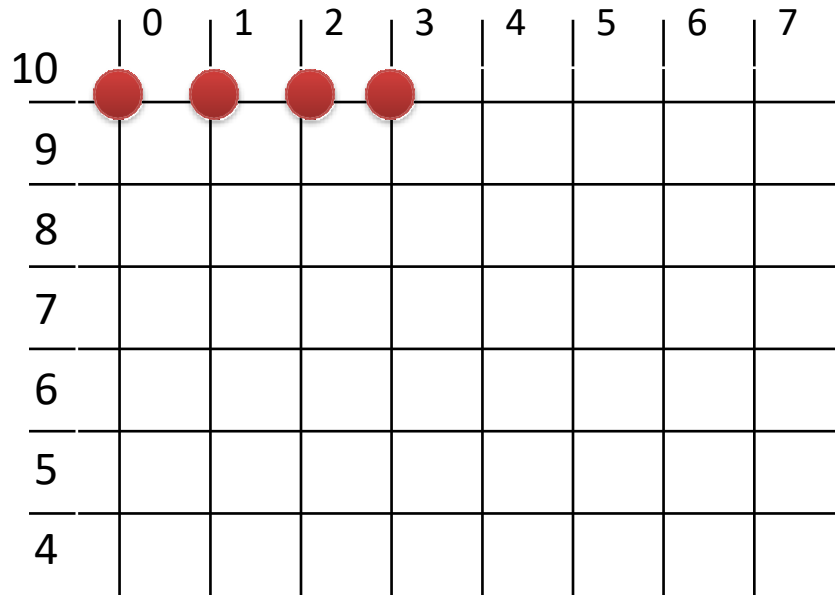
$(x,y) = (0,10)$

$h = 1 - R = -9$

K	1	2	3				
2x	0	2	4				
2y	20	20	20				
h	-6	↙ -1					
(x,y)	E(1,10)	E(2,10)	E(3,10)				

$h \leq 0, E$

Example



Given:

Radius , $R = 10$

$(x,y)=(0,10)$

$h = 1 - R = -9$

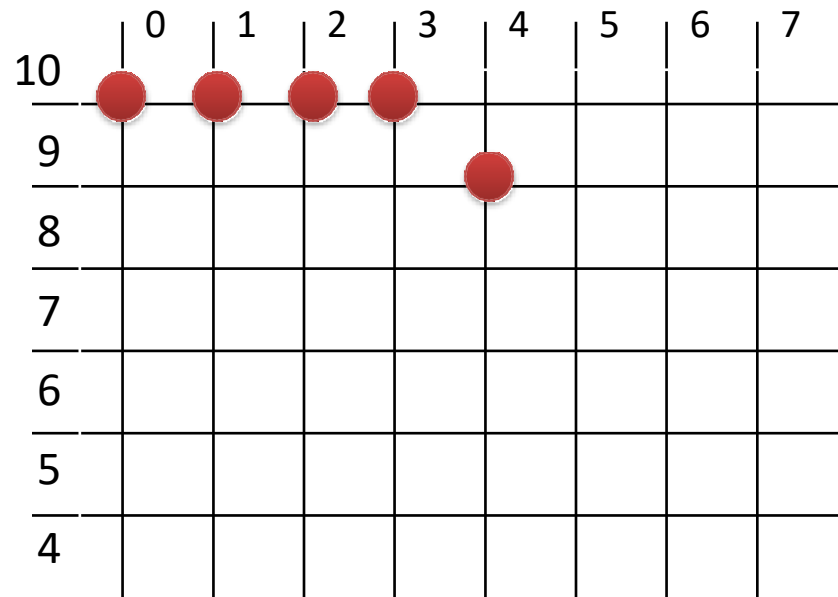
$h = h + \Delta E = h + 2x + 3$

$= -1 + 4 + 3$

$= 6$

K	1	2	3				
2x	0	2	4				
2y	20	20	20				
h	-6	-1	6				
(x,y)	E(1,10)	E(2,10)	E(3,10)				

Example



Given:

Radius , $R = 10$

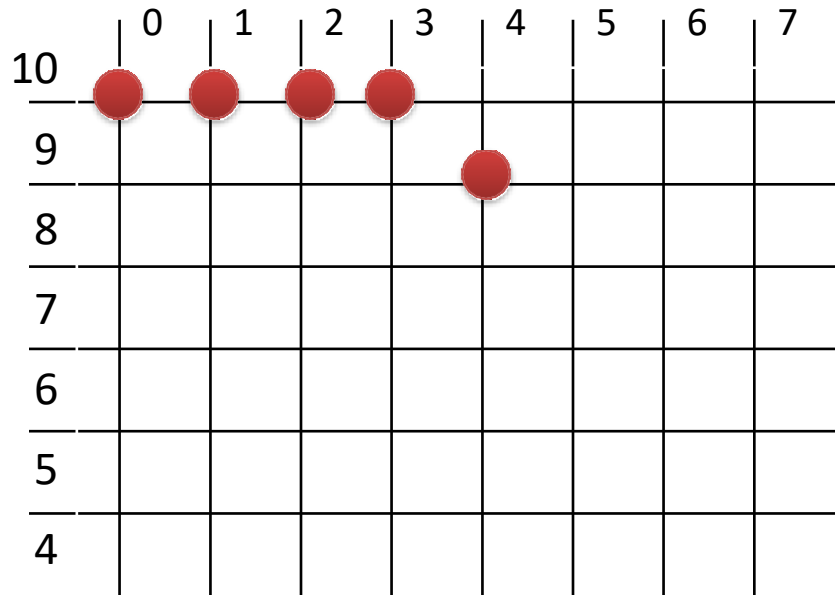
$(x,y) = (0,10)$

$h = 1 - R = -9$

K	1	2	3	4			
2x	0	2	4	6			
2y	20	20	20	20			
h	-6	-1	↓ 6				
(x,y)	E(1,10)	E(2,10)	E(3,10)	S(4,9)			

$h > 0, SE$

Example



Given:

Radius , $R = 10$

$(x,y)=(0,10)$

$h = 1 - R = -9$

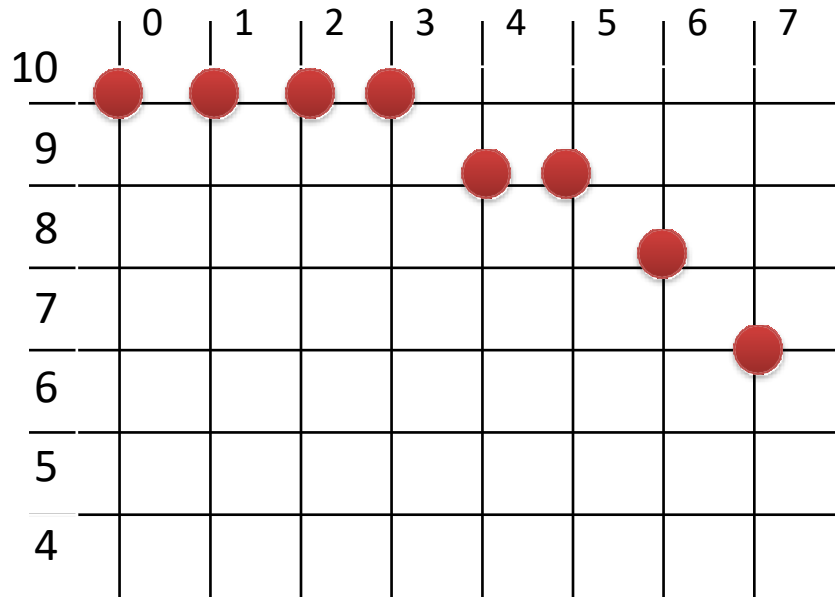
$h = h + \Delta SE = h + 2x - 2y + 5$

$= 6 + 6 - 20 + 5$

$= -3$

K	1	2	3	4			
2x	0	2	4	6			
2y	20	20	20	20			
h	-6	-1	6	-3			
(x,y)	E(1,10)	E(2,10)	E(3,10)	S(4,9)			

Example



Given:

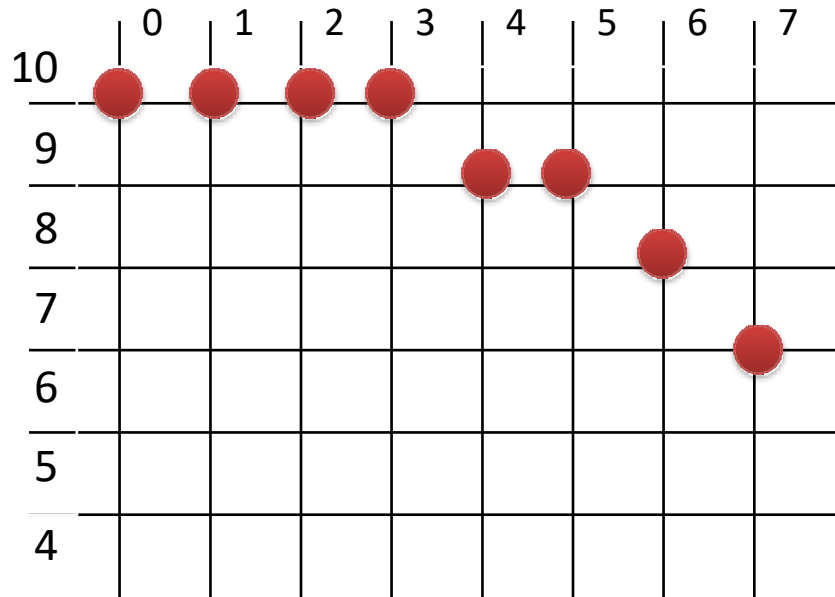
Radius , $R = 10$

$(x,y) = (0,10)$

$h = 1 - R = -9$

K	1	2	3	4	5	6	7
2x	0	2	4	6	8	10	12
2y	20	20	20	20	18	18	16
h	-6	-1	6	-3	8	5	6
(x,y)	E(1,10)	E(2,10)	E(3,10)	S(4,9)	E(5,9)	S(6,8)	S(7,7)

Example



Given:

Radius , $R = 10$

$(x,y) = (0,10)$

$h = 1 - R = -9$

Until $y > x$

K	1	2	3	4	5	6	7
2x	0	2	4	6	8	10	12
2y	20	20	20	20	18	18	16
h	-6	-1	6	-3	8	5	6
(x,y)	E(1,10)	E(2,10)	E(3,10)	S(4,9)	E(5,9)	S(6,8)	S(7,7)

Practice Problem

- Perform the midpoint algorithm to draw a circle's portion at 7th octant which has center at (2,-3) and a radius of 7 pixels. Show each iterations and plot the points.